

A manifest of docs/reports

One hundred and eleven independent mathematical research packages,
classified by subject

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1 Scope and conventions

This document catalogues the one hundred and eleven research packages stored under `docs/reports`. The collection originally held one hundred and four reports; after two consolidations it contributes one hundred and two maintained reports. The other nine came through ProveIt's incoming-reports procedure (`docs/incoming`) — six in batch 36 and one each in batches 40, 41 and 44 — whose archives remain in the repository's history. Later batches also extended existing reports; their entries name the batch. The next paragraphs describe the historical collection before those two consolidations. On 29 September 2026 the two Lipparini Problem 6.2 reports were unified, and the finite-height transfinite-word report was integrated with its guarded-block companion. Both proof routes and all distinct extensions are retained; old directories contain redirects, not extra catalogue entries. The seven surreal-number reports that this catalogue used to carry were moved out to a repository of their own, with their history, and are no longer listed here. They arrived as one hundred and twenty-five ZIP archives, in seven deliveries, which have been unpacked and grouped into the subject hierarchy described in Section 2. The archives themselves are no longer kept in the repository.

One hundred and twenty-five archives yield one hundred and three reports for two different reasons. Twenty archives duplicated another: sixteen pairs, three three-way clusters and one four-way cluster. Two more were folded into a neighbour not because they repeated it but because they shared a subject — the three reports on term counts in the derivatives of power towers are now one article with a common framework and three parts. That consolidation is editorial, and the distinction matters: those three were put through the duplicate sweep and cleared it, and their three theorems are three different theorems. The seventh delivery of eighteen contained no duplicate at all, which is worth recording: a sweep that finds nothing is evidence about the collection, not a sweep that failed. Each pair has been merged into one report that proves the shared theorem once and keeps everything both originals built on top of it. Where the two reached a result by genuinely different routes, both proofs are kept and marked as alternatives, with a sentence on what each one buys; the merged shuffle report keeps three independent finite certifications of one theorem, and the merged report on surreal option graphs kept two different five-vertex counterexamples for the same reason before it moved out with the other surreal-number reports. A merged entry below names both of its source archives and says what was doubled.

Not every pair aimed at one problem is a duplicate. The two Gonshor reports both prove his product-birthday bound, but on domains that are incomparable in both directions, so neither theorem contains the other and they are catalogued separately; the same holds for reports answering different numbered problems of one paper, noted where it occurs.

Every package is self-contained and follows the same shape: a typeset article (`*.pdf`) with its complete L^AT_EX source, a `README.md` summarising the result, usually a `code/` directory with exact-arithmetic verification scripts, a `data/` or `results/` directory holding the recorded output of those scripts, and often `STATUS.md`, `sources.md` or `notes/` files auditing the sources consulted and the precise limits of the claim.

Two cautions apply uniformly and are therefore not repeated in the individual entries. First, these are AI-assisted research drafts: none has been independently refereed, and none has been checked in a proof assistant. Several continue a Lean or Rocq project elsewhere in ProveIt; that

confers no formal status on them, and their entries say which project they continue. Second, the descriptions below record *what each report sets out to settle and what it claims to prove*. Compiling this manifest involved reading each package’s statement of its problem and result; it did not involve checking any proof. Where a report itself flags a limitation — a prior disproof, an imported theorem, a correction to the source, or an exception left open — that is noted, because it changes what the report is.

The naming of the directories is editorial. Several distinct reports attack the same published problem, and some arrived under colliding archive names, so each directory has been given a name describing its own angle on the problem; the merged directories usually name their combined scope. The September 29 consolidations retain the base directory names to preserve existing links. The original archive names are recorded with every entry.

2 The directory tree

Category	Subcategory	Reports
enumerative-combinatorics	—	22
ordinals-and-order-types	ordinal-arithmetic	3
	wqo-powersets-and-statures	4
	friendly-order-types	3
	games-on-ordinals	3
	transfinite-words	1
	(top level)	5
generating-functions-and-asymptotics	oeis-sequence-asymptotics	9
	binary-product-coefficient-counts	3
	(top level)	5
log-concavity-and-unimodality	—	12
hankel-determinants	catalan-and-ballot	4
	somos-and-elliptic	3
	growth-and-runs	2
	(top level)	1
congruences-and-valuations	supercongruences	4
	iterated-series	2
	(top level)	3
automata-and-formal-languages	—	7
tetration-and-digit-stabilization	—	6
graph-theory	—	3
quaternionic-analysis	—	1
jacobian-conjecture	—	3
galois-theory-and-radicals	—	2
Total		111

More than a quarter of the collection consists of counterexamples and corrections rather than proofs. Ordinal Chomp, open-query games, compositional tree series, the secant numbers, the decimal-length bound for tetration stabilization, power-tower stabilization, iterated-power digit stabilization, cluster variables, chromatic coefficients of cycles, products of q -integers, domination roots, Hardinian array constants, the mixed-base counterexample, recursive exponents, Gregory sign thresholds and binary substitution discrepancy all refute, or replace with a proved statement, an assertion published or catalogued elsewhere. Several reports whose main outcome is a proof still correct the statement they prove: the Somos-8 certificates, the shifted Catalan Hankel polynomials and the Rueppel determinants record a sign error, a missing one-unit shift and a numbering ambiguity in their sources, and the independence-system thresholds report refines its target after finding that the original conjecture had already been disproved by someone else.

3 Ordinals and order types

3.1 Friendly order types

Three reports on Isa Vialard’s *friendly order type* $f(P)$, the rank of the tree of open-ended bad sequences in which every chosen point has an incomparable partner in the current residual. All three prove the same finite formula; they differ in what they build on top of it, and in how they prove it — one by a forest-rank monovariant that survives moves deleting a whole upset, one

through a lemma bounding the cut maxima. A fourth report proved the formula by the first report's argument almost verbatim and has been merged into it.

1. Friendly Order Types Through Incomparability Components

[ordinals-and-order-types/friendly-order-types/incomparability-components-and-infinite-tails/friendly_order_types.pdf](#)
 from [friendly_order_type_research](#) (1).zip

The same component formula, together with an exact friendly rank for connected infinite well-partial-orders: writing $o(C) = \lambda + n$ with λ the limit part and $T(C)$ the finite set of points with finite principal upsets. Includes an exhaustive finite verification and a proof audit.

2. Friendly Order Type Through Incomparability Components

[ordinals-and-order-types/friendly-order-types/matroid-rank-and-ordinal-sums/article.pdf](#)
 from [friendly_order_type_research](#) (3).zip

Identifies $|P| - c(P)$ with the graphic-matroid rank and the incidence-matrix rank of the incomparability graph. Adds a substitution formula, a finite Cartesian-product formula, and additivity over arbitrary well-ordered ordinal sums; evaluates every well-partial-order whose incomparability components are finite, and gives countable counterexamples delimiting stronger infinite extrapolations.

3. Friendly Order Types of Finite Posets and Ordinal Products

[ordinals-and-order-types/friendly-order-types/components-forests-and-ordinal-products/friendly_order_types.pdf](#)
 from [friendly_order_types_research.zip](#) + [friendly_order_type_research.zip](#)

Proves $f(P) = |P| - c(\text{Inc}(P))$ for a finite poset P , where c counts connected components of the incomparability graph (isolated vertices included), and classifies the witnesses: every maximum friendly subset is P minus one minimal root per component, counted by $\prod |\text{Min}(C_i)|$. With Vialard's width-transfer theorem this gives ordinal width $\omega^{f(P)}$ for the Dershowitz–Manna multiset ordering. On the transfinite side it proves $f(\alpha_1 \times \cdots \times \alpha_r \times B) = \Theta$, the natural product, except that it is the predecessor of Θ when every α_i is a successor and B has a greatest element, and classifies arbitrary finite ordinal boxes. Also: spanning-forest certificates, exact monotonicity under added comparisons, a bivariate generating function, and explicit four-point factors with equal (o, h, w, f) whose products differ. *Merged; the Cartesian-product formula is proved twice, once by corner deletion and once by a chain-grid spanning-subgraph argument.*

3.2 Well-quasi-ordered powersets and statures

Four reports on the open questions in Section 6 of Abriola, Halfon, Lopez, Schmitz, Schnoebelen and Vialard, *Measuring well-quasi-ordered finitary powersets* (arXiv:2312.14587v2), and on the corresponding topological invariant.

4. Cofinal strata and ordinal absorption

[ordinals-and-order-types/wqo-powersets-and-statures/cofinal-strata-of-finitary-powersets/article.pdf](#)
 from [cofinal_strata_ordinal_powersets.zip](#); batch 42: [ProveIt_Ordinal_Heights_Matchings.zip](#);
 batch 36: [ProveIt_Nonuniform_Powerset_Heights.zip](#), [ProveIt_Persistent_Frontier_Heights.zip](#),
[persistent_frontier_heights.zip](#), [ProveIt_Ordinal_Heights_Research.zip](#)

For $S_\alpha(Q)$, a finite poset Q with each point replaced by a chain $\alpha = \omega^\rho$, proves $o(K(S_\alpha(Q))) = \sum_k \omega^{\rho \otimes k} c_k(Q)$ and $h(K(S_\alpha(Q))) = \alpha \cdot \text{height}(M(Q))$, where c_k counts size- k maximal antichains not dominated by a larger one. The maximal-order-type theorem allows nonuniform pure well-partial-order fibers and is proved by a finite cofinal-stratification argument. Parts II and III, merged in batch 36 from four manuscripts that each prove it, supply the nonuniform height formula the original left open. For positive pure fibers $\omega^{\rho q}$, $h(K)$ is the largest ordinary ordinal sum, along a chain of maximal frontiers, of the largest fiber leaving the frontier at each step plus the largest fiber of the last frontier; after Cantor-normal-form refinement the same formula

covers nonzero limit fibers. For arbitrary ordinal fibers, finite and successor ones included, the height is given instead by a finite dynamic program over downsets with fallback weight 1, and the frontier compression is not valid there. Adds exact point ranks, NP membership of the threshold problem for the explicit pure-block encoding, and a separate part for well-partial-order fibers with $h(K(P_q))$ a positive pure power. Part IV, added in batch 42, shows that for height-two skeletons with uniform fibers the height is $\alpha(\nu_{\text{ur}} + 1)$, with ν_{ur} the maximum uniquely restricted matching number, so deciding the height is NP-complete already when every fiber is ω .

5. Intrinsic ordinal approximations for finitary powersets

`ordinals-and-order-types/wqo-powersets-and-statures/
intrinsic-ordinal-approximations/article.pdf`
from `intrinsic_ordinal_approximations.zip`

Studies the intrinsic invariant $a_o(A) = \sup\{o(B) + 1 : \text{Idl}(B) \hookrightarrow A\}$. Proves the universal bound $a_o(A) \leq h(P_f(A))$ with equality exactly when every finitely generated downset of A is an algebraic dcpo, gives the closed formula $h(P_f(\prod_i \alpha_i)) = \sup(\bigotimes_i (1 + a_i) + 1)$, and shows the universal identity fails on $\lambda + F$.

6. Ordinal Order Maps and Finitary Powersets of Ordinal–Finite Grids

`ordinals-and-order-types/wqo-powersets-and-statures/
ordinal-order-maps-and-grids/ordinal_order_maps.pdf`
from `ordinal_order_maps_research.zip`

Exact maximal order types and heights for finite Hoare powersets of $\beta \times Q$ with Q a finite poset: for $\beta = \omega^{\gamma_0} + \dots + \omega^{\gamma_{m-1}}$, $o(\text{Mon}(Q, \beta)) = \bigoplus_{s \text{ isotone}} \omega^{\bigoplus_q \gamma_{s(q)}}$.

7. Sharp ordinal bounds for Hoare powerspaces

`ordinals-and-order-types/wqo-powersets-and-statures/
hoare-powerspace-statures/article.pdf`
from `hoare_statures_research.zip`

For all Noetherian spaces, $1 + \|X\| \leq \|HX\| \leq 2^{\|X\|}$, with witnesses attaining both endpoints at every ordinal and T_1 upper witnesses; the older ω^α bound is attainable at infinite input stature α exactly when $\omega \cdot \alpha = \alpha$; the complete spectrum at $\omega + n$ is $\omega \cdot j(U) + (j(Q) - j(U))$. Extends the known well-quasi-order results of Abriola et al. to arbitrary Noetherian spaces.

3.3 Transfinite words

One unified report on Harry Altman’s finite-alphabet sharpness question (arXiv:2409.03199v2, Example 3.15), combining three source manuscripts. The common height-two formula is stated once. The finite-height induction and the more detailed guarded-tail investigation are retained as distinct routes.

8. Finite-alphabet transfinite words: exact order types and guarded periodic blocks

`ordinals-and-order-types/transfinite-words/order-types-below-omega-squared/article.pdf`
from `omega2_order_types_proposed_solution.zip` + `ordinal_words_research.zip` +
`finite_alphabet_transfinite_words.zip`

Claims $o(V_k(P)) = \omega^{\omega^{N_k(P)-1}}$ for $|P| \geq 2$ and each positive finite k , using canonical finite atom posets with $N_{k+1} = |P| + J(I_k(P)) - 1$. At height two this is $\omega^{\omega^{n+j(P)-2}}$; empty and singleton alphabets are explicit exceptions. Retains the selected-tail-family classification, its universal-only exception, chain/antichain extremality, exact-length strata, and canonical linear-time token comparison. The guarded proof has both natural-product and elementary amplification routes; the finite-height proof uses protected separators. Two independent implementations and their separate audit records are preserved. Neither an all-height union formula nor a higher-cutoff selected-family classification is claimed.

3.4 Ordinal arithmetic

Three reports on ordinal and infinitary arithmetic, including a consolidation of two independent proposed solutions to Problem 6.2 of Paolo Lipparini, *Monotone infinitary operations on ordinals* (arXiv:2505.00424v2): define and study the least weakly monotone, e -special strictly increasing operation on countable sequences of ordinals.

9. An Explicit Formula for the Minimal e -Monotone Infinitary Ordinal Sum

[ordinals-and-order-types/ordinal-arithmetic/lipparini-minimal-infinitary-sum/article/explicit_ordinal_sum.pdf](#)

from `lipparini_problem_6_2_proposed_solution.zip` + `ordinal_operation_research.zip`

The same three-case formula for the least admissible operation is proved by two retained routes: profile ranks with arithmetic absorption and finite-product lower bounds, and corrected blocks with threshold interpolation and finite-part restoration. The latter avoids the product-rank lemma; both import Lipparini's characterization of the weaker S . Retains the multiset quotient, bounded-poset heights, finite-cap ranks with alternative proofs, equality on comparable inputs, cardinality preservation, and counterexamples to grouping and continuity. The correction is zero, a finite increment, or one *natural* summand ω . Both implementations and their separate verification records are preserved.

10. Jacobsthal Products All at Once

[ordinals-and-order-types/ordinal-arithmetic/jacobsthal-products-all-at-once/article.pdf](#)

from `Jacobsthal_Products_All_at_Once.zip`

Question 2.16 of Altman, *Intermediate arithmetic operations on ordinal numbers* (MLQ 63 (2017) 228–242), asks for an all-at-once order-theoretic model of finite and infinitary Jacobsthal products, rather than one built from parenthesized binary multiplication or a limit. Gives a constructive affirmative answer: a Cantor-normal-form comparison rule on actual tuples or finite-support functions, invariant under order isomorphisms and extending the componentwise order, whose definition never evaluates a Jacobsthal product. Adds a restricted maximality characterisation and a finite-tail reduction. The associativity identity and the binary normal-form formula are classical and are not claimed here; the construction is not claimed to be preserved by arbitrary order embeddings.

11. Pairwise Coherence Does Not Guarantee Completion

[ordinals-and-order-types/ordinal-arithmetic/](#)

[pairwise-coherence-of-infinitary-products/ordinal_product_coherence.pdf](#)

from `ordinal_product_coherence.zip`

Answers the pairwise-coherence sufficiency question of Lipparini, *Noncommutative infinitary semigroups* (arXiv:2605.28413v1, Remark 7.1(b)), negatively. On the two-element group C_2 , retaining every finite product and declaring every infinite all- e product to be e gives a partial system satisfying full pairwise regrouping, even for arbitrary partitions, yet with no injective extension defining all finite and ω -indexed products; two elements is minimal. Adds: for any monoid M the canonical partial domain $P(M)$ is coherent, and for directly finite M it completes exactly when the unit group is trivial; and for every r a system on $3r + 4$ elements coherent through nesting depth r but conflicting at depth $r + 1$, so no fixed finite depth of the coherence test characterises completability.

3.5 Games on ordinals

12. An ordinal Chomp transition at 2

[ordinals-and-order-types/games-on-ordinals/](#)

[ordinal-chomp-transition-at-two/ordinal_chomp_counterexample.pdf](#)

from `ordinal_chomp_transition_counterexample.zip` + `ordinal_chomp_counterexample.zip`

For a numerical semigroup S ordered by $x \leq_S y$ iff $y - x \in S$, and the ordinal extension ordered by ordinary ordinal subtraction (so S^2 is the *lexicographic*, not coordinatewise, square), Chomp on S

is a second-player win while Chomp on S^σ is a first-player win for every $\sigma \geq 2$, and $\omega \cdot 4 + 4$ is the *least* winning opening. Hence $\text{ch}(\langle 4, 5, 6 \rangle) = \text{ch}(\langle 4, 6, 9 \rangle) = 2$, a negative answer to Question 4.6 of Rivero Herrera (arXiv:2504.07317v3) and a refutation of Conjecture 4.1 of v1, already for three small natural-number generators. The proof reduces the infinite game to a finite spectral statement via a finite-window recurrence and an exact finite-state certificate, then transfers it by an explicit response procedure. *Merged; the shared $\langle 4, 6, 9 \rangle$ certificates were verified to hold the same 253 rows entry for entry, and the two reports' poisoned and unpoisoned conventions are reconciled by an explicit dictionary.*

13. Every ordinal as a point-separating game value

[ordinals-and-order-types/games-on-ordinals/
point-separating-game-values/ordinal_separation.pdf](#)
from ordinal_separation_research.zip

For every ordinal λ , orders $\{0, 1\}^\lambda$ lexicographically and gives it the topology of all downward-closed sets, and the refinement generated by downward-closed sets minus finite sets. The point-separating game value is exactly λ for both: the first space is compact T_0 , the second compact T_1 , and for infinite λ neither is Hausdorff. Taking $\lambda = \omega^2 + 1$ supplies a compact T_1 witness of cardinality continuum for Problem 1.2 of Chiozini, Csernák and Soukup (arXiv:2510.05754v3), with no CH, forcing or large-cardinal hypothesis. Hausdorff witnesses are not claimed. This is a different problem of that paper from the one the open-query games report answers.

14. Open-query games on ordinal spaces

[ordinals-and-order-types/games-on-ordinals/open-query-membership-games/article.pdf](#)
from ordinal_membership_games.zip; batch 52: ProveIt_Finite_Output_Open_Query_Games.zip

A negative answer to Problem 1.3 of Chiozini, Csernák and Soukup (arXiv:2510.05754v3): two copies of $[0, \omega]$ each have set-membership number 1, but their topological sum has value 2. The general theorem computes the invariant for every nonempty Hausdorff space of finite Cantor–Bendixson height h as $\lceil \log_2(h + 1) \rceil$ when the last nonempty derivative is a singleton and $\lceil \log_2(h + 2) \rceil$ otherwise. Part II, added in batch 52, replaces the membership bit by one of q output labels: the optimal depth is $\lceil \log_2 L \rceil$ for the least number L of ordered layers with open suffixes; on Hausdorff spaces of finite height $L \leq (q - 1)h + s$, sharply, with s the number of labels on the last derivative; a topological sum of components of depth at most d has depth at most $d + \lceil \log_2 q \rceil$, sharply; and computing the optimal depth is polynomial for two labels but NP-complete for three, even on disjoint chains. At $q = 2$ it reproduces Part I.

3.6 Order-theoretic invariants without a subcategory

Five reports whose objects sit outside every subcategory above: the minor order on finite graphs, two on sites of filters on ω , congruences of finite lattices, and a translation-invariant ideal.

15. Ordinal complexity of graph minors

[ordinals-and-order-types/graph-minor-ordinals/graph_minor_ordinals.pdf](#)
from graph_minor_ordinals.zip

For finite simple graphs of maximum degree at most two under minor containment, computes maximal order type, ordinal antichain-tree width and descending-tree height: $o(D) = w(D) = \omega^{\omega \cdot 2}$ and $h(D) = \omega$, with exact values ω^{b+k} and ω^{b+k-1} for the subfamilies bounded by path size b and cyclic-component count k . The general theorem is that a wqo C of connected graphs with $o(C) = \alpha < \varepsilon_0$ has finite disjoint-union closure of type ω^α , proved by a component-splitting argument, since minor containment is not multiset embedding. A corollary is that continuity fails for increasing unions of minor-closed classes. The report is explicit that it does not compute the maximal order type of the minor order on all finite graphs and settles no named conjecture.

16. A two-valued obstruction to gluing Fréchet powers

[ordinals-and-order-types/frechet-two-valued-obstruction/article.pdf](#)

from frechet_two_valued_obstruction.zip

For the presheaf $F \mapsto A^\omega / \sim_F$ on proper filters on ω containing the cofinite filter, with the dense Grothendieck topology, proves in ZF that it is a sheaf if and only if $|A| \leq 1$. The counterexample is an explicit matching family of constant sections using two values, on a covering sieve with countably many generators each of countable base; the distinction between the number of sieve generators and the size of the full sieve is essential to it. The question is Massas, *A Semi-Constructive Approach to the Hyperreal Line* (2023), p. 517 after Fact 5.10, repeated in the 2024 dissertation *Duality and Infinity*, p. 249.

17. Bounded-character Fréchet filter sites

ordinals-and-order-types/frechet-bounded-character-sites/article.pdf

from filter_complexity_research.zip, Frechet_Cardinal_Boundaries.zip

Continues the previous entry's non-claim about other filter sites. For the site P_κ of free filters on ω generated by fewer than κ sets modulo the cofinite filter, represents the Booleanization C_κ as a regular-open algebra on the free ultrafilters and proves that $P(\omega)/\text{Fin}$ is order dense in it exactly for $\kappa \leq \mathfrak{p}$; above $\mathfrak{p}$ no complete Boolean homomorphism preserving the modulo-finite events exists in either direction. The atoms are the ultrafilters of character below κ , so C_κ is atomless exactly for $\kappa \leq \mathfrak{u}$; $\mathfrak{h}$ is the first size of a family of events that cannot be decided jointly on a dense set of regions. Computes the associated sheaf by Boolean mixtures, with full Boolean-valued transfer, countable saturation of ultrafilter specializations, and sizes $\mathfrak{c}$, $2^{\mathfrak{c}}$, $2^{2^{\mathfrak{c}}}$ for the raw power and the two sheafifications. A three-stage example uses Millán's forcing model as an imported input. Merged from two manuscripts whose indexings differ by a successor shift; the second one's closure-based results are stated at successor stages only.

18. A sharp cover bound for congruences of finite lattices

ordinals-and-order-types/lattice-congruence-cover-bound/sharp_cover_bound.pdf

from lattice_cover_bound_research.zip

If a finite n -element lattice has an element with at least $k \geq 3$ upper covers then $|\text{Con}(L)| \leq 2^{n-k-1}$, with equality exactly for the glued sums $C_s \oplus M_k \oplus C_t$. This replaces the multicolour Ramsey threshold $R_{k-1}(k)$ of the source's Lemma 4.4 (arXiv:2603.11454) by k itself. Adds a sharp nonextremal gap, a fan-label ideal refinement, and a bound on the size of Czédli's skeleton in terms of congruence density. All congruence machinery, including Freese's maximum, is proved from scratch. The target is an informal suggestion in the source, not a numbered conjecture.

19. A non-Baire translation-invariant ideal

ordinals-and-order-types/non-baire-translation-invariant-ideal/article.pdf

from non_baire_ideal_research.zip

An affirmative construction for Question 1 of Filipów and Tryba, *Densities for sets of natural numbers vanishing on a given family* (J. Number Theory 211 (2020) 371–382; arXiv:2309.00982, p. 7): a translation-invariant ideal that is not Baire and carries a perfect translation-almost-disjoint family.

4 Automata and formal languages

Five reports on state complexity, synchronization and language hierarchies. The first two are each a merge: the shuffle bound arrived four times and the DFAO reversal bound twice.

20. The six-state shuffle bound

automata-and-formal-languages/shuffle-six-state-bound/shuffle_six_state.pdf

from shuffle_six_state_research (1).zip + shuffle_six_state_research.zip +
Shuffle_Six_State_Theorem.zip + six_state_shuffle_research.zip

For complete DFAs with $m, n \geq 2$ and $\min(m, n) \leq 6$, the worst-case deterministic state complexity of the shuffle is $F(m, n) = 2^{mn-1} + 2^{(m-1)(n-1)}(2^{m-1} - 1)(2^{n-1} - 1)$, over a finite alphabet allowed to depend on m and n ; the unrestricted conjecture stays open. The route is a reachability theorem

in the universal shuffle automaton — a subset is reachable exactly when it meets the initial row and the initial column — proved by induction on $(m+n, |S|)$ down to a finite certificate obligation. *Merged from four reports. Two shipped the same 10,642 witnesses; a third supplies an independent database of 10,619 witnesses sharing no witness with them, so the finite step is certified twice over; a fourth adds a fifth analytic reduction that cuts the obligation to 453 certificates and replaces the solver search with a deterministic enumeration. All three routes are kept.*

21. A coloring obstruction to maximal binary DFAO reversal

[automata-and-formal-languages/dfao-reversal-coloring-obstruction/article.pdf](#)
 from dfao_reversal_research.zip + binary_dfao_reversal_research.zip; batch 39:
 ProveIt_Three_Output_Reversal.zip; batches 40–41: ProveIt_DFAO_Eventual_Optimality.zip,
 ProveIt_Binary_DFAO_Reversal.zip, ProveIt_Binary_Reversal_Exactness.zip; batch 43:
 ProveIt_Four_Output_Reversal.zip, ProveIt_Four_Output_Reversal (1).zip

For transformations a, b on an n -element set and an output map to k labels, proves $|\tau\langle a, b \rangle| \leq k^n - k! + g(k) < k^n$ for $3 \leq k \leq n$, with $g(k)$ the largest order of a permutation of k symbols — the binary-alphabet nonattainment question of Davies, arXiv:1705.07150v2, Section 5. The mechanism is a collision-orbit graph, a rightmost singular factorization, and a cyclic relabeling bound. Adds an abelian-permutation extension, a structural universal bound matching the published lower construction on all 372 pairs with $7 \leq n \leq 30$, and polynomial-time missing-coloring certificates. *Merged; the two originals ran the same three-case argument, and both a qualitative short proof and the counting proof are kept.* Part II, added in batch 39, settles three outputs exactly: $R_2(n, 3) = 3^n - 3(2^A + 2^B - 2) + B$ for every $n \geq 7$, with (A, B) the closest coprime split of n , together with rigidity of the extremizers; the formula matches all twenty-four $k = 3$ rows of the finite table. Part III, merged in batches 40–41 from three manuscripts, settles every $k \geq 4$ past an explicit threshold: $R_2(n, k) = k^n - \text{Bip}_k(a_n, b_n) + a_n b_n$ once $n \geq N(k) \leq 3k^4$, and for all $n \geq \max(7, k+1)$ when $k = 4, \dots, 7$ (computer-assisted below $N(k)$), with the lower construction imported from Davies; a non-effective eventual route for every $k \geq 3$, 919 finite certificates and a second proof of the $k = 3$ formula are kept. Explicit exactness for $k \geq 8$ at small n stays open. Part IV, merged in batch 43 from two manuscripts on $k = 4$, re-derives that case with an analytic threshold $n \geq 26$ (against 212), and adds stability, the missing-orbit inventory, the extremal output maps and a minimal order-15 recurrence.

22. Synchronizing Dzyga’s automata by a cyclic retract

[automata-and-formal-languages/dzyga-synchronizing-retract/article.pdf](#)
 from Dzyga_Automata_Cyclic_Retract_Research.zip

For the Figure-3-complete automaton A_k of Conjecture 8 in Szykula’s 2026 survey, exhibits an explicit word $R_k = p(ec^{r-1})^{r-2}e$ with $r = k + 4$ that sends every state to q_2 for every $k \geq 1$, of expanded length $8k^2 + 54k + 92 = \frac{8}{9}n^2 + \frac{22}{3}n + 16$ for $n = 3k + 6$. The proof is a cyclic retraction with two projected reflections whose discrepancy supplies one adjacent merger, and does not depend on the finite computations; shortening the reflected tail by one state gives a nonsynchronizing automaton with a dihedral permutation quotient. This settles the synchronization clause of the stated model, not the numerical threshold conjecture, and the universal lower bound $\tau(q_0) \geq 4k + 8$ is not proved.

23. Gaps in bounded-shuffle hierarchies

[automata-and-formal-languages/insertion-degree-spectra/article.pdf](#)
 from shuffle_spectra_research.zip; batch 41: ProveIt_Binary_Insertion_Spectra.zip

A counterexample to the insertion-degree No-Gap Conjecture of Hughes, arXiv:2608.27755v1, Section 11, followed by positive results: every finite set of positive integers containing 1 is the insertion-degree spectrum of two nonempty finite languages over a suitable alphabet, and any prescribed finite list of degrees above 1 is realizable with every other output of degree 1. A ternary pair has spectrum $\{1, 3\}$ with sole difficult output **abcba**, whose length 5 is the minimum possible for a gap witness; a binary pair has spectrum $\{1, 2, 4\}$ while all 18 singleton source-pair spectra are initial intervals, so source-pair ambiguity can erase a degree. The separate iteration-depth no-gap conjecture is proved for arbitrary languages. Part II, added in batch 41, fixes the alphabet:

every finite set containing 1 is the spectrum of finite *binary* languages with the single inserted word a^m , so two letters always suffice; its m^2 length bound holds in its full-coverage class only.

24. Constrained crossover closure

[automata-and-formal-languages/constrained-crossover-closure/article.pdf](#)
from `ProveIt_Constrained_Crossover.zip` (batch 44)

Answers the question Hughes (arXiv:2608.27755v1, Section 4) leaves open on the complexity of deciding $L \otimes_c L = L$ for the constrained crossover of a context-free language: it is co-r.e.-complete. Adds an aligned-rank no-gap theorem for iterated crossover and effectivity results in which Presburger quantifier elimination is used only as an algorithmic step. Related to, but using nothing from, the insertion-degree report above.

25. A cut-sensitive attack on the first open MIX grammar case

[automata-and-formal-languages/mix-grammar-cut-sensitive/mix_grammar_report.pdf](#)
from `mix_grammar_research.zip`

The selected question — whether $L_3 = \text{MIX}$ for the canonical tuple grammars of Yuyama, *On the Kanazawa–Salvati Conjecture* (arXiv:2609.13871v1) — is *not* settled here, and the report says so at the front. What it does contain, with complete written proofs, is a cut-sensitive membership recurrence, a broad three-component positive family, an infinite family in $L_3 \setminus L_2$, and an exhaustive computation through length 21.

26. Natural densities of the Tribonacci word’s additive complexities

[automata-and-formal-languages/tribonacci-additive-complexity/article.pdf](#)
from `tribonacci_additive_density_research.zip`

Answers Remark 19 of Popoli, Shallit and Stipulanti (arXiv:2410.02409). With β the Tribonacci constant, the additive complexity of the Tribonacci word takes values in $\{3, 4, 5\}$ with natural densities $d_3 = (21\beta^2 - 38\beta + 5)/22$, $d_4 = (-55\beta^2 + 72\beta + 67)/22$ and $d_5 = (17\beta^2 - 17\beta - 25)/11$, all positive and summing to 1, with a power-saving error $O(N^\theta)$, $\theta = \log(7/5)/\log \beta$, at *every* integer cutoff rather than only along Tribonacci cutoffs. The 76-state output automaton is reconstructed independently rather than copied, and certified by evaluating its degree-76 characteristic polynomial at the adjacency matrix.

5 Hankel determinants and continued fractions

5.1 Somos sequences and elliptic curves

27. A proof of Barry’s Hankel–Somos-6 conjecture

[hankel-determinants/somos-and-elliptic/barry-somos-6-hankel-recurrence/article.pdf](#)
from `barry_somos6_research.zip`

Conjecture 7 of Paul Barry, arXiv:2211.12637: for the Hankel determinants H_N of a four-parameter Jacobi-type generating function, $H_N H_{N-6} = \alpha H_{N-1} H_{N-5} + \gamma H_{N-3}^2$ with $\alpha = t^2(r+2)^2$ and $\gamma = t^3((s+t)(r+s+t+2)^2 + t(r+2)^3)$. Also proves a homogeneous four-parameter form and a seven-parameter polynomial Hankel theorem making exceptional specialisations rigorous; the verifier never divides by a determinant.

28. Finite Hankel Certificates for Three Somos-8 Conjectures

[hankel-determinants/somos-and-elliptic/somos-8-hankel-certificates/article.pdf](#)
from `somos_hankel_proofs.zip`

Conjectures 11, 12 and 13 of the same Barry paper, by importing Hone’s genus-two Hankel theorem (arXiv:1907.05204v3, Thm. 5.4) and supplying a binomial translation, a polynomial-minor specialisation valid on the degenerate locus, and a rank-bound lemma that turns four symbolic rows into an all-index identity. Identifies a sign error in the source’s Catalan expression for Conjecture 13.

29. Elliptic addition and Jacobi continued fractions

[hankel-determinants/somos-and-elliptic/
elliptic-addition-and-jacobi-fractions/article.pdf](#)
from [elliptic_continued_fraction_research.zip](#)

A consistently indexed, non-torsion version of Barry's 2023 Conjecture 4, with the coordinate formulas of Conjecture 3: every continued-fraction tail is given explicitly and removing one layer is shown to be elliptic translation. Hypotheses: characteristic zero, $Y^2 + aXY + bY = X^3 + cX^2 + dX$ nonsingular, $b \neq 0$, and $P = (0, 0)$ of infinite order; order-three and order-four examples show the regular fraction genuinely needs the last one.

5.2 Catalan and ballot moments

30. Shifted Catalan Hankel Polynomials

[hankel-determinants/catalan-and-ballot/shifted-catalan-hankel-polynomials/article.pdf](#)
from [catalan_hankel_conjecture_solution.zip](#); batch 39: [ProveIt_Catalan_Root_Collisions.zip](#);
batch 42: [ProveIt_Cyclotomic_Catalan_Recurrences.zip](#); batch 53:
[Two_Endpoint_Catalan_Confluence.zip](#); batch 55: [ProveIt_Interior_Collision_Research.zip](#)

For $D_N^{(m)}(a, b) = \det(aC_{i+j+m} + bC_{i+j+m+1})$, proves the denominator-exponent assertion on page 4 of Barry, arXiv:2011.10827v1, and shows the separately printed Conjecture 2 needs an explicit one-unit shift correction. The minimal characteristic polynomial is $(X^2 - (a+2b)X + b^2)^{m(m-1)/2+1}$, so the minimal order is $m(m-1) + 2$; all exceptional reduced denominators, the exact coefficient formula, numerator degree and reciprocal symmetry are supplied. Part II, added in batch 39, replaces $aC + bC'$ by an arbitrary polynomial multiplier and computes the root collisions: exact sector degrees with nonzero leading constants, the minimal recurrence order off resonance, a secondary-Hankel factorization and a certified root-free algorithm for the recurrence order. Part III, added in batch 42, gives every characteristic multiplicity for one repeated root at a cyclotomic resonance, and shows that the question it answers needed one more parameter: the sign of z^L , not only t and $\text{ord}(z^2)$. Part IV, added in batch 53, answers the endpoint half of Part II's question on uniform limits: for fixed-size clusters of roots approaching the endpoints 0 and 4 (d roots in all) the normalized determinant is a convergent series in even powers of $1/(N + d/2)$, locally uniform through every collision inside a cluster, with a factorized leading profile, an explicit first interaction term, and ratio to the fixed-endpoint determinant tending to 1 exactly when the outward displacements are $o(N^{-2})$. Part V, added in batch 55, answers the other half: for fixed-size clusters of roots colliding on the scale $1/N$ at fixed bases other than 0 and 4, the determinant is an exact finite sum of exponential sectors whose amplitudes are convergent series in $1/(N + d/2)$, uniform through every collision, with an explicit first correction (this expansion is not even), an explicit remainder bound, and factorized amplitudes for several separated clusters. For two equal-multiplicity roots inside the spectrum the limit is an oscillatory Jacobi Hankel determinant, the first correction oscillates as $\sin(2(N+t)\theta)$, and simple zeros move by a convergent phase-dependent expansion. Limits uniform up to the endpoints remain open.

31. Product formulas for ballot-polynomial Hankel determinants

[hankel-determinants/catalan-and-ballot/
ballot-polynomial-hankel-determinants/article.pdf](#)
from [Cigler_Hankel_Conjectures_13_15.zip](#)

Conjectures 13, 14 and 15 in Section 5 of Johann Cigler, arXiv:2111.14492v3. An explicit coefficient product gives the sign and exact support of the normalised shift- k Hankel determinant of the ballot moments and positivity of every residual coefficient for $n \geq 1$; the same products give the conjectured generating-function denominators, exact numerator degrees and reciprocal numerator identities. Adds strict log-concavity of the residual coefficients and their limiting binomial profile.

32. Rectangular Schur polynomials and Cigler's Conjecture 8

[hankel-determinants/catalan-and-ballot/cigler-conjecture-8-schur/article.pdf](#)

from `cigler_conjecture8_research.zip`

A complete proof of Conjecture 8 in Cigler, arXiv:2111.14492v3, Section 4 (printed pp. 16–17), for the Hankel determinants of $b_m(t) = \sum_r \binom{\lfloor m/2 \rfloor}{r} \binom{\lceil m/2 \rceil}{r} t^r$, including the numerator-polynomial degree and reciprocity assertions. A different conjecture of that paper from the ballot-polynomial entry (Conjectures 13–15) and from the parity entry (Conjecture 16).

33. A parity factorization for Cigler’s Conjecture 16

`hankel-determinants/catalan-and-ballot/cigler-conjecture-16-parity/article.pdf`

from `cigler_conjecture16_research.zip`

A complete proof of Conjecture 16 in Cigler, arXiv:2111.14492v3, Section 6 (pp. 21–22), on the Hankel polynomials, together with a sharp stabilization statement: the report also determines the first coefficient at which the conjectured stabilization fails. Its own README explains why this is a different problem from the other Cigler entries in this collection.

5.3 Arithmetic Hankel determinants

The numerator formula recorded as a conjecture in OEIS A122251, with A122249 as its binary special case. Two reports attacked it and proved it the same way, sharing even their worked counterexamples; they are merged here into one. The merged report sits at the category’s top level.

34. A Proof of an OEIS Hankel-Numerator Conjecture

`hankel-determinants/a122251-numerators-and-denominators/report.pdf`

from `hankel_numerator_research.zip` + `hankel_oeis_conjecture_proof.zip`

For the Hankel determinant

$$H_n(m, a) = \det(1/(a + m(i + j)))_{0 \leq i, j \leq n},$$

the reduced numerator is $\prod_{p|m} p^{n(n+1)v_p(m)+2 \sum_{k \leq n} v_p(k!)}$: independent of the coprime offset a , a perfect square, and supported on the primes dividing m . Taking $m = p$ and $a = 1$ settles the OEIS A122251 conjecture, with A122249 as the binary case. Also gives a closed form for the reduced denominator and all its local valuations, a sharp gcd theorem over the allowed offsets realised at just two explicitly constructed shifts, the extension to unequal row and column steps $\det(1/(a + m_1 i + m_2 j))$, recurrences, digit-sum asymptotics, and a Lambert-series generating function with a natural boundary. *Merged; the determinant is evaluated three independent ways, and a second proof of the numerator formula runs through an explicit inverse matrix and MacMahon’s boxed plane partitions.*

5.4 Growth laws and run statistics

35. A proof of the Apéry Hankel growth conjecture

`hankel-determinants/growth-and-runs/`

`apery-hankel-determinant-growth/apery_hankel_growth.pdf`

from `apery_hankel_research.zip`

For $A_k = \sum_j \binom{k}{j}^2 \binom{k+j}{j}^2$ and $D_n = \det(A_{i+j})$, proves $\log D_n = n(n+1) \log \frac{17+12\sqrt{2}}{4} + O(n)$ — the limit conjectured in the September 2026 revision of OEIS A228143 — using Edgar’s moment-density theorem. Also an explicit law for $\det(A_{m(i+j)+r_n})$ with $r_n/n \rightarrow \rho$, and non-D-finiteness of the formal generating series of D_n .

36. Binary Runs in Rueppel Hankel Determinants

`hankel-determinants/growth-and-runs/rueppel-binary-run-determinants/rueppel_hankel.pdf`

from `rueppel_hankel_research.zip`

For the Rueppel series $r(x) = \sum_{j \geq 0} x^{2^j-1}$, the polynomial identity $H_n(tx + 1/r(x^2)) = (-1)^{1+\varepsilon+\lfloor R(n)/2 \rfloor} (\varepsilon + (R(n) - \varepsilon)t^2)$, where $R(n)$ counts runs of equal bits in the binary expansion of n and ε is the low bit of the Gray code $n \oplus \lfloor n/2 \rfloor$. Settles statements 8, 10, 13 and 23 of Barry’s 2021 paper (not the identically numbered ones elsewhere), with a correction.

6 Congruences and p -adic valuations

6.1 Supercongruences

37. Congruences for OEIS A348410 at every prime

congruences-and-valuations/supercongruences/a348410-all-prime-congruences/article.pdf
from a348410_supercongruence.zip + A348410_Supercongruence_Research.zip

For $A_{\alpha,\beta}(n) = [x^n](1-x)^{-\alpha n}(1+x)^{-\beta n}$ proves the odd-prime cubic supercongruence $v_p(A(Np) - A(N)) \geq 3v_p(N) - \varepsilon_p$ with $\varepsilon_3 = 1$, $\varepsilon_p = 0$ for $p \geq 5$, and no coprimality condition; $(\alpha, \beta) = (2, 1)$ is Bala's A348410 conjecture. Separately proves an exact binary leading-defect formula, $A(m2^r) - A(m2^{r-1}) \equiv 2^{3r-3} \alpha \beta \binom{\alpha+\beta+1}{m-1}^{m-1} \pmod{2^{3r-2}}$, hence the all-prime bound $3r - v_p(12)$ and sharp denominators 12 and 24. An appendix audits arXiv:2104.10754v1, whose framing conclusion and weighted harmonic assertion fail on its own displayed examples. *Merged; the odd-prime and binary halves run on different mechanisms — a doubling automorphism, and reflection with formal integration by parts — and both are kept.*

Not a duplicate of *signed-binomial-supercongruences*, and checked. That report's $a_{r,s,t}(n) = [x^{tn}](1+x)^{rn}(1-x)^{sn}$ specializes at $t = 1$, $(r, s) = (-\beta, -\alpha)$ to the coefficient studied here, and the two odd-prime statements then agree. Neither contains the other. That one is general in the extraction step t , which is fixed at 1 here; this one adds the prime 2 with an exact defect and its sharpness, every prime at once, minimal common denominators, the primitive-cyclic-word interpretation, the Lambert/Euler product, the algebraic generating functions, a spacing-three counterexample, and a sufficient criterion beyond the family. The proofs are different arguments: that one imports Jacobsthal's congruence, this one uses none and pairs a quadratic-tail estimate with a coefficient estimate.

38. Signed binomial supercongruences and OEIS A352373

congruences-and-valuations/supercongruences/
a352373-signed-binomial-supercongruences/signed_binomial_supercongruence.pdf
from OEIS_A352373_supercongruence.zip

For $a(n) = [x^{tn}](1+x)^{rn}(1-x)^{sn}$ proves $a(mp^h) \equiv a(mp^{h-1}) \pmod{p^{3h}}$ for $p \geq 5$ and $a(m3^h) \equiv a(m3^{h-1}) \pmod{3^{3h-1}}$, with m not required coprime to p . Covers A352373, A348410, A351856 and A351857 as specialisations; the only non-elementary import is the classical Jacobsthal binomial congruence.

Not a duplicate of *all-prime-congruences*, and checked. Setting $t = 1$ and $(r, s) = (-\beta, -\alpha)$ turns the theorem below into that report's odd-prime tower, and the moduli agree after reindexing the multiplier. Neither contains the other: this one is general in the extraction step t , that one is fixed at $t = 1$ but proves the prime 2 case, the all-prime bound, the sharp denominators and much else. The two proofs are genuinely different — this one imports Jacobsthal's congruence, that one does not use it at all.

39. Apéry numbers under exponentiation and series reversion

congruences-and-valuations/supercongruences/
a267220-apery-transform-supercongruences/article.pdf
from Apery_OEIS_Supercongruences.zip, apery_oeis_conjectures(1).zip

Proves both supercongruence conjectures contributed by Peter Bala to OEIS A267220 on 17 October 2024, for every integer parameter including negative and zero. With $A(x) = \exp \sum_{n \geq 1} a_n x^n / n$ for the Apéry numbers a_n , $F(x) = x / \text{Rev}(xA(x))$, $u_m(n) = [x^n]A(x)^{mn}$ and $v_m(n) = [x^n]F(x)^{mn}$: for odd p and $R = \nu_p(N)$ both families satisfy $\nu_p(u_m(N) - u_m(N/p)) \geq 2R + \nu_p(m)$, the unsigned statement for v_m holds at *every* prime including 2, and a *signed* refinement $(-1)^{mN} u_m(N) \equiv (-1)^{mN/p} u_m(N/p)$ recovers full precision at $p = 2$, where the unsigned bound genuinely loses one power. Merged from two reports that share a seed lemma and a Möbius/dilogarithm layer but diverge at the crux; both routes are kept, and the merged text is careful that the signed and unsigned 2-adic statements are different statements, not a sharper and a weaker version of one.

40. Four Motzkin congruences

[congruences-and-valuations/supercongruences/a001006-motzkin-congruences/article.pdf](#)
from [motzkin_congruences.zip](#)

Proves all four conjectures in Peter Bala's comment of 10 February 2022 on OEIS A001006, with sharp ranges. The engine is a prime-power block-transfer theorem: for odd p and $q = p^{k-s+1}$, $h = mp^{s-1}$, the Motzkin numbers satisfy $M_{mp^k+d} \equiv T_h M_d$ for $0 \leq d \leq q-3$, with explicit corrections at $d = q-2$ and $d = q-1$; the boundary case gives $M_{np^{r-2}} \equiv \frac{\varepsilon_p^r-1}{2} T_{n-1} - U_{n-1} \pmod{p}$. Adds the exact first failures beyond the two initial-block ranges, arbitrary-multiplier and all-prime-power boundary extensions, a prime-modulus digit algorithm, and examples showing why unweighted repetition modulo a prime cube is generally false. Discusses Burns, Kohen and Pan-Sun rather than claiming novelty.

6.2 Congruences for iterated and compositional series

41. Arithmetic of a Self-Iterating Generating Function

[congruences-and-valuations/iterated-series/a396807-self-iterating-series/article.pdf](#)
from [A396807_congruences_and_prime_power_lifts.zip](#)

Both conjectures in OEIS A396807 for $A(x) = x + A^{[5]}(x)A^{[6]}(x)$ (functional iteration): $A(x) \equiv x/(1-x)$ and $A^{[k]}(x) \equiv x/(1-kx)$ modulo 10 for every integer k , negative iterates included. Adds a sharp modulus theorem for $F = x + cF^{[r]}F^{[s]}$, rational lifts at every selected prime power, and minimal period 46860 for A396807 modulo 100 starting at index 1.

42. Prime-power congruences for compositional tree series

[congruences-and-valuations/iterated-series/compositional-tree-series-congruences/article.pdf](#)
from [compositional_tree_congruences.zip](#)

For $A_l(x) = x \exp(A_l^{ol})$, shows the OEIS A396805 claim $a_5(n) \equiv n \pmod{5}$ is *false*: first failure $n = 23$, with exception set $\{20j+3, 20j+4 : j \geq 1\}$ and residue 2 throughout. Proves eleven other conjectural assertions in A396803–A396806 via a general reduction of coefficients mod p^α to typed rooted trees on fewer than p^α labels, yielding eventual periods, rational residue generating functions, and an exact classification of $a_l(n) \equiv n \pmod{p}$.

6.3 Valuations, divisibility and periodicity

43. Binary Carries and the Divisibility of Two-Stack-Sortable Permutation Numbers

[congruences-and-valuations/a000139-binary-carries-parity/research.pdf](#)
from [a000139_carries_research.zip](#)

Proves Peter Bala's parity conjecture for OEIS A000139, $a(n) = 2(3n)!/((n+1)!(2n+1)!)$: $a(n)$ is odd exactly when n is odd and its binary expansion contains no 11. The full valuation is $v_2(a(n)) = s(n) + s(n+1) - s(3n)$ in binary digit sums. Two independent parity proofs, plus exact valuation distributions, recurrences, asymptotics, extremal formulas and limit laws.

44. Polynomial divisibility of height-one factorial ratios

[congruences-and-valuations/factorial-ratio-polynomial-divisibility/article.pdf](#)
from [bala_factorial_divisibility_research.zip](#)

A general rational-root criterion for uniform polynomial divisibility of balanced integral factorial ratios, applied to Peter Bala's uniform-product conjectures for OEIS A211417 and A295431: explicit multipliers for every product length, with the least multipliers certified in eight particular examples. Includes an atlas of the 52 sporadic height-one ratios with recurrences, generating functions and asymptotic expansions.

45. When the Secant Numbers Become Periodic

[congruences-and-valuations/secant-number-periodicity/article.pdf](#)

from secant_periodicity_research.zip

Refutes the OEIS A000364 conjecture that the secant numbers from $n = 1$ are purely periodic modulo every m : at $m = 27$, $a(1) \equiv 1$ but $a(19) \equiv 10$. The exact replacement is that pure periodicity from index one holds if and only if no odd prime cube divides m ; the least eventual period always divides $\varphi(m)$. Determines the least eventual period and exact onset for every modulus, with densities and a certified mean onset.

7 Tetration and digit stabilization

46. A Counterexample to a Decimal-Length Bound for Tetration Stabilization

[tetration-and-digit-stabilization/congruence-speed-decimal-length-bound/article.pdf](#)

from tetration_counterexample.zip

An explicit 2544-digit base $A = 1 + qu$ ($q = 5^{2547}$) for which $\nu_{10}(T_{n+1} - T_n) = \min(1 + 2548n, 2547(n + 1))$ at every height, so the congruence speed at height $\text{len}(A) + 2$ is 2548 while the permanent value is 2547. This contradicts the decimal-length form of Conjecture 1 of Marco Ripà, *The congruence speed formula* (NNTDM 27(4), 2021). The report audits the later Ripà-Onnis confirmation: their weaker valuation bound is compatible with the counterexample, the decimal-length bound is not.

47. Tetration, perfect powers, and frozen decimal digits

[tetration-and-digit-stabilization/perfect-power-minima-for-frozen-digits/article.pdf](#)

from tetration_perfect_power_minima.zip

Determines $\rho(n)$, the smallest n th perfect power $x^n > 1$ whose eventual number of newly frozen terminal decimal digits is exactly n : $\rho(1) = 2$, $\rho(2) = 49$, and $\rho(n) = B(s)^n$ for $n \geq 3$ with $s = n - v_2(n)$ and $B(s) \in \{2^s \pm 1, 3 \cdot 2^s \pm 1\}$ according to $s \bmod 4$. All other roots are ruled out by a deterministic quadratic size barrier plus small exact certificates.

48. Phase-shift classification for integer tetration

[tetration-and-digit-stabilization/](#)

[phase-shift-classification/tetration_phase_classification.pdf](#)

from tetration_phase_classification.zip

Proves the 21-word phase-shift conjecture in OEIS A376842, the 14-word even-base restriction from the appendix of Ripà's 2025 paper, and the 28-word prescribed-anchor conjecture in OEIS A376446. The mechanism compares the 2-adic and 5-adic valuations of $D_b = T_{b+1} - T_b$ at the first permanently constant-speed height: strict 2-adic dominance gives phase 5, strict 5-adic dominance a multiplicative orbit in $\mathbb{F}_5^\times$, and balanced valuations exactly four further anchored words.

49. A Sharp Convergence Theorem for the q -Newton Series of Subcritical Tetration

[tetration-and-digit-stabilization/q-newton-series-convergence/article/](#)

[tetration_qnewton.pdf](#)

from tetration_qnewton_research.zip

Proves convergence of the explicit q -binomial interpolation series of MathOverflow question 259278 for every real base $1 < a < e^{1/e}$, with $\log a = qe^{-q}$: the grouped series converges absolutely and locally uniformly for $\text{Re } z > -2$, converges conditionally on $\text{Re } z = -2$ except at $z = -2 + 2\pi ik/(-\log q)$, and diverges for $\text{Re } z < -2$. Adds strict complete monotonicity and uniqueness under an eventual log-convexity condition.

50. Stabilization of integer power towers

[tetration-and-digit-stabilization/power-tower-stabilization/article.pdf](#)

from tetration_stabilization_research.zip + tetration_exact_digits.zip

Refutes the claim that every prime base ending in 9 reaches its permanent decimal congruence speed by height 2 (Ripà, Conjecture 2; arXiv:2208.02622v1, Conjecture 4.1): 2749 is the smallest prime counterexample, with $H(2749) = 3$. Proves exact formulas for $C_a(h)$, $V(a)$ and $H(a)$, an exact natural density for onsets, and infinitely many primes at each prescribed onset via CRT

and Dirichlet. Also resolves the three conjectural comments in OEIS A324017 — two proved, the third disproved, with $3 \uparrow 2 \equiv 27$ and $3 \uparrow 3 \equiv 59 \pmod{64}$ — via $T_{h+r} - T_h \equiv -2q^h \pmod{q^{h+1}}$ for even $q \geq 4$, so exactly one radix- q digit freezes per height. Adds a factorization-free digit-lifting algorithm, modular saturation across Knuth arrow ranks, and a sign-sensitive local p -adic Lambert W formula. *Merged; the shared one-digit-per-height theorem was proved identically in both originals and is proved once here.*

51. Delayed digit stabilization: a counterexample and sharp bounds

tetration-and-digit-stabilization/
iterated-power-digit-stabilization/delayed_digit_stabilization.pdf
from digit_stabilization_research.zip

For $S_a(b) = v_{10}(a^{10^{b+1}} - a^{10^b})$ and $D_a(b) = S_a(b) - S_a(b-1)$, answers negatively the question whether $D_a(b) = 1$ for all $b \geq 3$: the smallest counterexample is $a = 3 \cdot 2^{99} + 1$, with $S_a(2) = 100$, $S_a(3) = 102$, $D_a(3) = 2$. Proves minimality, an exact onset formula, the smallest counterexample at every stage, the natural density of exceptions, and arbitrary-radix extensions. The article is explicit that a^{10^b} is not genuine tetration and derives the tower formulas separately.

8 Log-concavity, unimodality and positivity

52. Single curves do not force log-concavity

log-concavity-and-unimodality/cluster-variable-log-concavity-counterexample/article.pdf
from annular_single_lamination_counterexample.zip

An unpunctured annulus with one marked point on each boundary and a single unit-weight simple lamination curve gives, after mutations 2, 1, 2 with the two mutable variables specialised to 1, the polynomial $1 + 3q^2 + 2q^3 + 4q^4 + 2q^5 + q^6$; the consecutive coefficients 3, 2, 4 form a strict valley, refuting Conjecture 1 of arXiv:2508.04396v3 and conflicting with its general unpunctured-surface unimodality statement. The geometric realisation is proved, not merely the matrix calculation, and a fixed seed gives infinitely many such examples.

53. Cycles refute infinite log-concavity for chromatic coefficients

log-concavity-and-unimodality/infinite-log-concavity-of-cycles/article.pdf
from chromatic_cycle_logconcavity.zip

For the cycle C_n with absolute chromatic coefficients under the iterated operator $L(a)_k = a_k^2 - a_{k-1}a_{k+1}$: infinitely log-concave for $3 \leq n \leq 11$; first failure at depth 5 for $n = 12$, depth 4 for $13 \leq n \leq 16$, and depth 3 for $n \geq 17$. So C_{12} is the smallest cycle counterexample (no minimum over all graphs is claimed), with an explicit third-iterate witness polynomial valid for all $n \geq 3$.

54. Oscillating Log-Concavity in Geometric Generating Functions

log-concavity-and-unimodality/oscillating-log-concavity-geometric-transform/article.pdf
from oscillating_log_concavity_research.zip

Challenges 2 and 3 in Section 3 of Heim and Neuhauser, arXiv:2302.13327v1, for the *geometric* transform $1/(1 - uG(z))$. For divisor-sum weights, every real exponent and every $u > 0$ give infinitely many strict log-concavity and log-convexity indices with eventually bounded gaps; likewise for integer power weights with exponent ≥ 2 , the exponents 0 and 1 being explicitly exceptional. For square weights at $u = 1$, every four consecutive indices carry both signs (sharp), each with density $1/2$ in every arithmetic progression. A subdominant-pole theorem is the common mechanism.

55. A counterexample and a sharp spacing-three criterion for products of q -integers

log-concavity-and-unimodality/q-integer-product-unimodality/report.pdf
from q_unimodality_attack.zip

The necessity half of Conjecture 5.4 of Connelly, Ito, Martinez, Shevchenko and Yang (arXiv:2605.12822v1) is false: $(1+q)^6(1+q^3)$ is symmetric and weakly unimodal although no a_i is divisible by 3 and $b = 2$ exceeds the asserted bound, and the example lies inside the source's

claimed range. The report also proves the proposed sufficient condition for every $r \geq 2$, a necessary degree bound in the absence of an r -divisible ordinary factor, and a complete classification of spacings two and three.

56. Sharp size thresholds for failures of log-concavity in independence systems

[log-concavity-and-unimodality/independence-system-thresholds/article.pdf](#)

from [sharp_log_concavity_thresholds.zip](#)

For each rank $r \geq 4$, determines the smallest ambient ground-set size admitting failure of ordinary, ordered and ultra log-concavity, and shows the three thresholds coincide across all independence systems, those satisfying Xie–Xu’s augmentation bound, and those with additive augmentation parameter 3 (values $7/6/5$, $7/7/6$, $8/7/7$, and $r + 1$ throughout for $r \geq 7$). The initial target, Xie–Xu Conjecture 4.8 (arXiv:2408.09152v1), had already been disproved by Alex Chengyu Li in September 2026; the report states this and refines the target to the rankwise minimum-size question.

57. A sharp order-five threshold for higher-order Stirling subset log-concavity

[log-concavity-and-unimodality/higher-order-stirling-rows/article.pdf](#)

from [stirling_log_concavity.zip](#)

For $S_{n,k}^{(r)}$, the number of partitions of $n + (r - 1)k$ labels into k blocks of size $\geq r$, proves that all rows are log-concave if and only if $1 \leq r \leq 5$, with strict inequalities in the interior — the log-concavity assertion of Deb–Sokal Conjecture 1.4(c) (arXiv:2507.18959v1). Also proves the cycle cases $r = 3, 4$; the fifth-order *cycle* conjecture is explicitly left unresolved, and the included operator counterexample is flagged as not being a counterexample to it.

58. A sum-of-squares proof of Ulas’s real-zero conjecture

[log-concavity-and-unimodality/stern-polynomial-positivity/stern_conjecture_note.pdf](#)

from [stern_conjecture_research.zip](#)

Conjecture 4.4 of Maciej Ulas, arXiv:1909.10844v1, for the Stern polynomials indexed by $a_n = (4^n - 1)/3$ and $h_n = 2(4^n - 1)(2 \cdot 4^n + 1)/3 + 1$: for every $n \geq 0$ and every real t , $W_n(t) = (A_{n+1}(t) - tA_n(t)/2)^2 + 3t^2 A_n(t)^2/4 > 0$. The adjacent-polynomial quadratic identity behind it was already in Ulas’s paper and is re-derived, not claimed as new.

59. Two-coin counterexamples to generalized Shepp–Olkin thresholds

[log-concavity-and-unimodality/bernoulli-entropy-quasiconcavity/article.pdf](#)

from [bernoulli_entropy_research.zip](#); batch 43: [proveit_product_entropy.zip](#)

For every finite real $q > 1$, both the Rényi- q and the Tsallis- q entropy of a sum of independent Bernoulli variables fail even *quasiconcavity* as a function of the parameter vector, already on the open square $(0, 1)^2$ with two coins. This disproves the proposed thresholds — Rényi 2 and Tsallis 3.65986... — adjoined to Conjecture 4.2 of arXiv:1503.01570 and restated in arXiv:1510.05390, Section 8. Part II, added in batch 43, settles the subunit joint-concavity question for the joint bit vector only: product Rényi entropy is concave in every dimension for $0 < q \leq 2$, and product Tsallis entropy is concave for every $0 < q < 1$ exactly when $n \leq 9$. For the ordinary sum the question stays open; an external claim about it is recorded as unverified.

60. Nonreal roots survive in polynomial iterative equations

[log-concavity-and-unimodality/nonreal-roots-in-iterative-equations/article.pdf](#)

from [nonreal_roots_research.zip](#); batch 41: [ProveIt_Increasing_Spectral_Core.zip](#); batch 42: [ProveIt_Orientation_Reversing_Spectra.zip](#); batch 54: [ProveIt_Exact_Algebraic_Degree.zip](#)

Here $P[f] = \sum_j a_j f^{oj}$ means a combination of *iterates*, not a polynomial in pointwise values. Gives a negative answer to Problem 6.1 of Draga–Morawiec (arXiv:1503.00570), which proposed deleting nonreal maximal-modulus characteristic roots: an explicit odd, strictly increasing, globally bi-Lipschitz homeomorphism realises a cubic whose nonreal root cannot be discarded. Adds a classification of the circular root spectra that occur. Part II, added in batch 41, classifies the minimal iterative polynomials of increasing homeomorphisms with roots on several circles, through a universal spectral core and an affine rigidity theorem. Part III, added in batch 42, does

the same for decreasing homeomorphisms with $P(0) \neq 0$: two-cycle and one-sided spectra, a decreasing core with one or two witnesses, and a finite-smoothness classification. Part IV, added in batch 54, answers Part III's question on algebraic degree: for a polynomial coordinate H and $f = H \circ (a \text{ id}) \circ H^{-1}$ the degree of f over $\mathbb{R}(x)$ is $\deg H / \deg W$, where W is the largest normalized right compositional factor of H with $W(at) = a^{\deg W} W(t)$, and it is exactly $\deg H$ for every coordinate of Part III's finite-smoothness theorem. For every odd $n \geq 5$ it gives real-analytic involutions of algebraic degree n , with distortion arbitrarily close to 1, whose graph extension has Galois group S_n , so they are not expressible by radicals.

61. Canonical binomial decomposition does not preserve real roots

[log-concavity-and-unimodality/binomial-decomposition-real-roots/article.pdf](#)

from `mu_welker_recursive_decomposition.zip` + `recursive_binomial_counterexample.zip`

A negative answer to Question 3.10 of Mu and Welker (arXiv:2503.24076v1) for their canonical decomposition $F(t) = G(t) + tH(t)$ built from the integer binomial expansion of every coefficient: real-rootedness of F does not pass to the parts. Two cubic witnesses are carried, one with a repeated root and one with simple roots, each with its own digit table, discriminant pair and exact nonreal zeros; the report also gives the minimal witness within cubics, two all-degree thresholds neither of which implies the other, and a two-sided limit. *Merged; the two originals were two write-ups of one proof program, each already containing both witnesses with the headline and the co-star swapped.*

62. A first-triple criterion for log-concavity of MDS weight distributions

[log-concavity-and-unimodality/mds-weight-log-concavity/article.pdf](#)

from `mds_log_concavity_research.zip`

For $d = n - k + 1 \geq 2$, $k \geq 3$ and $q > d$, the compressed formal MDS weight distribution $(1, A_d, \dots, A_n)$ is log-concave if and only if the single inequality $A_{d+1}^2 \geq A_d A_{d+2}$ holds — the whole sequence is controlled by its first triple.

63. A uniform Hankel obstruction for higher-order Stirling subset polynomials

[log-concavity-and-unimodality/stirling-hankel-obstruction/article.pdf](#)

from `stirling_hankel_obstruction.zip`

Proves the higher-order negative assertion of Deb–Sokal Conjecture 1.4(d), with a stronger statement at every positive shift: for $r \geq 3$ and $a \geq 1$ the 3×3 Hankel determinant $\det(s_{r,a+i+j}(x))$ of the row-generating polynomials fails the required positivity. Clause (d) of that conjecture; clause (c) is the subject of the higher-order Stirling rows entry.

9 Quaternionic analysis

One report, merged from ten manuscripts. The subject is new to this collection, and the delivery was unusual: one expository survey and nine independent attempts at the same follow-up problem, all written on the same day. The nine agree, which is the substance of the entry.

64. Slice regularity and the global inverse Fueter problem

[quaternionic-analysis/slice-regularity-and-fueter-inversion/article.pdf](#)

from `quaternionic_analysis.zip` and nine companion archives; batch 39:

`ProveIt_Fueter_Polynomial_Periods.zip`

One article assembled from ten manuscripts. Part I is the delivered expository survey: slice regularity and Cauchy–Fueter regularity side by side, with Cauchy formulas, Taylor and Laurent expansions, the regular product and the geometry of zeros, the argument principle, Rouché, open mappings, Schwarz–Pick and an intrinsic Riemann mapping theorem. Parts II–XI answer the follow-up question the other nine manuscripts all attacked independently: for an axially monogenic $g = P + IQ$, when is there a single-valued slice-regular f with $\Delta_4 f = g$? The answer is a two-period obstruction. The closed one-forms $\omega_g = \frac{1}{2}(-P dx + Q dy)$ and $\theta_g = \frac{1}{2}((xP + yQ)dx + (yP - xQ)dy)$ have vanishing periods exactly when a global primitive exists, and then $f(x + Iy) = (x + Iy)C + V$ with $dC = \omega_g$, $dV = \theta_g$, unique modulo $q \mapsto qa + b$; the monodromy is affine, and the cokernel of

the Fueter map has real dimension $8m$. The point the nine exist to make is that θ_g is computed from the target, with no first primitive chosen. **Nine independent derivations reaching the same two forms, the same $8m$ and the same logarithmic energy constant is the evidence the entry rests on.** Where they differed it was almost always notation — thirteen conflicts are recorded and resolved in `MERGE_NOTES.md` — but two were mathematical and would have produced a false statement if the merge had been mechanical: the size law at a critical singularity has genuinely different sharp constants in two norms that four manuscripts both called M_g , and the conserved flux recovering the slope residue needs the weight $3(\operatorname{Re} q - u) + \operatorname{Im} q$ rather than $q - u$, the coefficient 3 being exactly what makes it right-Fueter-regular. All ten source manuscripts and all ten verification programs are kept; the deduplication is in the article. Parts XII–XIII, added in batch 39 from an eleventh manuscript, answer the report’s open problem on higher dimensions in large part: in ambient dimension $2h + 2$ one polynomial closed form of degree $2h - 1$ (at $h = 1$ the form $\theta_g + t\omega_g$) gives a complete obstruction, with a constructive inverse, a cokernel of dimension $2h d_V m$, an exact residue filtration and sharp size and energy constants.

10 Graph theory

65. Mutual-visibility sets in Andrásfai graphs

[graph-theory/andrasfai-mutual-visibility/article.pdf](#)

from `andrasfai_visibility_research.zip`

Proves the generating function attributed to Andrew Howroyd in OEIS A391632, $\sum a(n)x^n = x(4 - 19x + 33x^2 - 40x^3 + 8x^4)/(1 - 10x + 29x^2 - 32x^3 + 8x^4)$, by an exact weight-preserving bijection between mutual-visibility sets of the n -Andrásfai graph and anchored closed walks of length $3n - 1$ in a four-state labelled graph. Adds the full bivariate generating function, an exact binomial formula for every cardinality, maximum-set classification, eventual polynomiality at each fixed deficit, exponential asymptotics, a central limit theorem, and linear-time recognition and exact uniform sampling.

66. A bicyclic 30-vertex graph with domination root -4

[graph-theory/domination-root-minus-four-order-30/domination_root_30.pdf](#)

from `domination_root_30_research.zip`

An explicit connected planar graph on 30 vertices and 31 edges, treewidth 2 and domination number 10, with $D(G, x) = x^{10}(x + 4)Q_{19}(x)$ — a negative answer to question 2 of Section 4 in Alikhani and Griswold, arXiv:2608.00109v1, which asks whether 33 is the minimum order. Their 33-vertex example had already disproved the original integer-root conjecture; no claim of global minimality at 30 is made. Three independent exact polynomial computations agree.

67. A marked-cycle proof of the trapezohedral domination conjecture

[graph-theory/trapezohedral-minimal-dominating-sets/article.pdf](#)

from `trapezohedral_domination_research.zip`, `A381190_proof_package.zip`

Proves the generating function labelled conjectural in OEIS A381190: for the n -trapezohedral graph, the number of dominating sets that are inclusion-minimal among all dominating sets and induce a connected subgraph is $a_n = 2n(2p_{n-2} + p_{n-3})$ with $p_m = p_{m-2} + p_{m-3}$ (Padovan). The proof classifies such sets by a cyclic arrangement of gaps of length 2 and 3 with a restricted marked edge. Adds the size distribution, strict log-concavity, an unbiased sampler, a Binet formula and a central limit theorem.

11 Enumerative combinatorics

68. A Twelve-Value Proof of the A181280 Formula

[enumerative-combinatorics/a181280-binary-matrix-formula/article.pdf](#)

from A181280_research_package.zip

The binary-matrix formula recorded as Conjecture 20 by Kauers and Koutschan (2023) and still marked conjectural in OEIS A181280: an explicit spectral recurrence bound and a finite certificate resting on twelve decisive counts, together with further consequences. STATUS.md records that general C-finiteness was already proved in Kauers’s December 2025 lecture slides.

69. Esthetic Numbers, Folded Paths, and Algebraic Diagonals

enumerative-combinatorics/esthetic-numbers-and-folded-paths/article.pdf

from esthetic_walks_research.zip

The diagonal conjecture $T(q, q - 2) = A182555(q - 2)$ in OEIS A377000, where $T(q, k)$ counts base- q positive integers of length k whose adjacent digits differ by one. Proves it exactly, proves the entry’s other two conjectures as retrieved, and derives formulas for every fixed-offset diagonal $q = k + d$.

70. Diagonal Hardinian Arrays: A Fixed-Parameter Asymptotic Formula

enumerative-combinatorics/hardinian-array-diagonal-asymptotics/article.pdf

from hardinian_arrays_research.zip

For every fixed $r \geq 0$, $H_r(n, n) \sim C_r 4^{rn} n^{-r(r-1)/2}$ with an explicit C_r in Gamma values. The formula confirms the constants tabulated by Dougherty-Bliss and Kauers (2024) but refutes their proposed restriction to powers of 2, 3 and π : $C_7 = 2^{18} 5^2 / (3^{45} \pi^3)$. Adds an exact finite determinant at fixed array size, an exact enumeration algorithm, and a limiting negative-binomial law for total deficits.

71. An explicit formula for lucky parking spots

enumerative-combinatorics/lucky-parking-spots/article.pdf

from lucky_parking_spots_research.zip

Proves Conjecture 13 of Butler, Hadaway, Lenius, Martens and Moats (JIS 29 (2026), Article 26.1.1): a closed form for $C(n, j)$, the number of parking functions in which *spot* j is lucky. Adds an exact reflection law, all fixed right-edge columns, a closed form for OEIS A374533, Poisson edge limits, uniform bulk convergence, polynomial reciprocity and asymptotic expansions.

72. A hexagonal-lattice proof of the A195806 conjecture

enumerative-combinatorics/a195806-hexagonal-lattice/article.pdf

from A195806_hexagonal_proof.zip

An elementary derivation of the six residue-class formulas and the constant-coefficient recurrence recorded as conjectural in OEIS A195806, starting from the original triangular-array constraints rather than from interpolation. The steps are an integral six-parameter bijection, four independent translation intervals, a dihedral chamber reduction, and an exact lattice count.

73. Leaves of large type in the numerical-semigroup tree

enumerative-combinatorics/numerical-semigroup-leaf-types/article.pdf

from leaf_type_counterexample.zip

Disproves Conjecture 4.4 of Chappelon, Ramírez Alfonsín and Stamate, *Numerical semigroup tree: type-representation* (arXiv:2507.15006v2), with an explicit genus-34 counterexample, and establishes a sharp asymptotic order for the type of leaves in that tree.

74. Strict growth and the Catalan limit for adjacency-bounded 132-avoiding permutations

enumerative-combinatorics/adjacency-bounded-132-avoiders/article.pdf

from strict_growth_132.zip; batch 38: Closing_Catalan_Growth_Gap.zip; batch 43:

ProveIt_Largest_Jump_Research.zip; batch 52: ProveIt_Macroscopic_Jump.zip,

ProveIt_Macroscopic_Jump_Research.zip

For 132-avoiding permutations with every adjacent difference at most m and growth constant α_m : proves $\alpha_{m+1} > \alpha_m$ for all $m \geq 1$ — the remaining strict-inequality part of Nadler’s Conjecture 2 — and the refinement $4 - \alpha_m = (2 \log m + 4 \log \log m + O(1))/m$, with $2 \log \pi - 4 \log 2 \leq \liminf E_m \leq \limsup E_m \leq 2 \log \pi$. Part II, added in batch 38, closes that window: a Catalan-weighted

supersolution and a growing cutoff give $E_m \rightarrow 2 \log \pi$, with an all-orders inverse-log expansion. Its numerics at $m = 10^6$ (near 3.15) are far from the limit and are not evidence for it. The all- m component ordering is still not claimed. Part III, added in batch 43, describes the joint limit near the boundary $m = n - d$: the distance $n - M_n$ of the largest jump converges in total variation at rate $n^{-1/2}$ to an explicit algebraic law, uniformly in $d = o(n)$. Part IV, merged in batch 52 from two manuscripts, treats the bulk regime $n - m \asymp n$: $\sqrt{n} P(n - M_n > un) \rightarrow (3/\sqrt{\pi})(1 - 2u)/\sqrt{u(1 - u)}$ for $0 < u < 1/2$ (and $\rightarrow 0$ for $u \geq 1/2$), so for $1/2 < \theta < 1$ the fraction of 132-avoiders of length n with every adjacent difference at most $\lfloor \theta n \rfloor$ is asymptotic to $3(2\theta - 1)/\sqrt{\pi\theta(1 - \theta)n}$, and the equivalent $3/\sqrt{\pi(1 - \theta)n}$, which Part III declined to assert, is false. It also gives the exact constant of every moment $E(n - M_n)^p$, $p > 1/2$ (for instance $E(n - M_n) \sim 3\sqrt{n/\pi}$); for $\theta \leq 1/2$ only $o(n^{-1/2})$ and polynomial lower bounds are proved.

75. Throwback Dynamics with Repeated Entries

[enumerative-combinatorics/throwback-queue-dynamics/throwback_research.pdf](#)
from [throwback_research.zip](#)

For the queue process that removes the head and reinserts it at the zero-based position equal to its weight: every token recurs if and only if at most m input tokens have weight at most m , for every $m \geq 1$, with an explicit description of the recurrent tokens when that fails. Adds exact cycle periods and frequencies, a coprimality criterion for perfect mixing, Catalan and prime-parking counts for recurrent cores, and the largest labelled period of a core of size r .

76. Reciprocity and Matrix Duality for Preorder Polytopes

[enumerative-combinatorics/preorder-polytope-reciprocity/article.pdf](#)
from [preorder_reciprocity.zip](#)

Proves Conjectures 7.2, 4.2 and 4.4 of Athanasiadis and Chapoton, *Polytopes and posets associated to preorders* (arXiv:2605.26916v1), and the ordinary $q = 1$ case of their Conjecture 4.9 (not its full q -refinement). Two mechanisms: Ehrhart–Macdonald reciprocity after subtracting the all-ones vector, extending the double identity to independently weighted capacities; and duplication of both colour classes of the bipartite graph followed by Postnikov’s lattice-point formula, giving the conjectured matrix transpose.

77. Support reciprocity and the q -zeta polynomial of preorder lattice-point posets

[enumerative-combinatorics/preorder-q-zeta-reciprocity/article.pdf](#)
from [preorder_qzeta_reciprocity.zip](#) + [preorder_q-zeta_reciprocity.zip](#) +
[preorder_q-reciprocity_research.zip](#)

Proves the full q -refinement of Athanasiadis–Chapoton Conjecture 4.9, equation (21): for a preorder τ on $[n]$, with P_τ the nonnegative integer vectors satisfying $a(I) \leq |I|$ on every order ideal and B_τ its maximal elements, $Z_q(P_\tau, [-1]_q) = (-1)^n \sum_{b \in B_\tau} q^{-|\text{supp}(b)|}$, together with the stronger identity at each exact support. The $q = 1$ case was already covered by the preorder-polytope reciprocity report and is cited, not reclaimed. *Merged from three reports that reached the same two theorems by three architecturally independent routes — an incidence inversion with Boolean localization, the same with a Hall-based privatization and a self-contained Euler identity, and a capacity-enumerator reciprocity with a transfer involution — and all three survive.*

78. An explicit shelling for preorder lattice-point posets

[enumerative-combinatorics/preorder-shellability/article.pdf](#)
from [preorder_shellability_research.zip](#)

An affirmative solution to Question 4.6 of Athanasiadis and Chapoton (arXiv:2605.26916v1, p. 13): an explicit pure shelling of the order complex of P_τ with its minimum removed, and a coning that shells the order complex of P_τ itself. Both complexes are therefore Cohen–Macaulay over the integers and over every field. A different question of that paper from the three other preorder entries.

79. Multiple-chain preorder polytopes

[enumerative-combinatorics/multiple-chain-exponential-formula/article.pdf](#)

from multiple_chain_exponential_formula_research.zip + multiple_chains_research.zip

Proves the exponential generating-function identity proposed in Section 9.5 of Athanasiadis and Chapoton (arXiv:2605.26916v1) for lattice points of multiple-chain preorder polytopes counted by number of nonzero coordinates, in a rectangular generalization with independent block width and height. Adds a duality corollary, an Ehrhart evaluation, a degree- 2^d root theorem with a quartic elimination certificate, two treatments of the γ -vector, a Stieltjes moment representation, the boundary constant in two forms, and a central limit theorem with exact variance. *Merged; the identity is proved twice, once through a subset encoding with a bounded symbol and once through slack meanders in a Laurent ring, and the eleven deliberate redundancies are listed in the text.*

80. Rationality and growth of stabilized permutation-weight series

enumerative-combinatorics/stabilized-permutation-weights/article.pdf

from stabilized_permutation_weights.zip

Studies the full stabilized generating functions of the permutation-weight q -Eulerian polynomials, beyond their initial connection with OEIS A256193: an all- d finite-spectrum theorem, rationality, a universal integer denominator, the exact radius of convergence and residue-class asymptotics. The two-descent sequence gets an explicit formula in powers of 3, powers of 2 and Fibonacci numbers.

81. A Reflexive Root-Polytope Model for Preorder h -Polynomials

enumerative-combinatorics/preorder-root-polytopes/article.pdf

from preorder_root_polytopes.zip; batch 39: ProveIt_Height_Two_Preorder_Research.zip; batch 42: ProveIt_Cactus_Rigidity.zip; batch 44: ProveIt_Preorder_Gamma.zip, ProveIt_Preorder_Gamma_Research.zip; batch 55: ProveIt_Lorentzian_Support.zip, ProveIt_Support_Polynomials_Research.zip, ProveIt_ADE_Stability.zip, ProveIt_Matroid_Lifts.zip

Proves Conjecture 5.1(c) of Athanasiadis and Chapoton (arXiv:2605.26916v1): for a preorder τ on $[n]$ the directed root polytope $C_\tau = \text{conv}(\{\pm e_i\} \cup \{e_i - e_j : j \leq_\tau i\})$ is reflexive with $h^*(C_\tau, t) = h_\tau(t)$, and any pulling triangulation of its boundary is combinatorially the boundary of a simplicial polytope with that h -polynomial. Palindromicity and unimodality, Conjecture 5.1(a,b), follow, as do the full g -theorem conditions. The higher-dimensional augmented bipartite identity and preorder duality are credited to Dai, Hou, Liu, Thawinrak and Wang (arXiv:2608.16037v2). Part II, added in batch 39, treats height-two preorders through matching polynomials: γ -positivity in height two (Conjecture 5.2 there), a family showing that real-rootedness (Conjecture 5.3) fails — its first counterexample is credited to a MathDB posting by Shivam Patel — and crown obstructions: an induced crown forces large minimal nonfaces in every lattice triangulation of C_τ , so the boundary complex is not flag when the crown has at least six elements. Flag realizability in general (Conjecture 5.1(d)) remains open. Part III, added in batch 42, shows that exactly the cacti admit a faithful unit-phase minor model, a property strictly stronger than real stability. Part IV, merged in batch 44 from two manuscripts, proves Conjecture 5.2 for every finite preorder: γ_k counts ordered pairs of disjoint k -sets joined by a τ -increasing bijection, and a reflexive relation gives a palindromic polynomial exactly when it is transitive. Part V, merged in batch 55 from two manuscripts, shows that for a bipartite graph with shores of sizes m and n every support enumerator divided by $\binom{\min(m,n)}{k}$ is log-concave (the unweighted inequality is Theorem A of Röhrle and Ulirsch, arXiv:2402.15317, carried through the support-counting bridge), so every h_τ and every height-two γ -vector is ultra-log-concave; it classifies the palindromic support polynomials of all bipartite graphs, gives FPRASes for partition functions and for single coefficients, and shows exact counting $\#P$ -complete already in height two. Part VI, added in batch 55 with the stability sections of a further manuscript, shows that stability of the signed support polynomial is block-local, holds for the incidence-apex graph of a tree exactly when the tree is a finite or affine ADE tree, and is classified for books and bipartite multi-theta graphs; the books show it strictly stronger than real-rootedness. Part VII, added in batch 55, shows that the transversal matroid of a preorder determines the preorder up to isomorphism and that height-two preorder matroids generate exactly the gammoids under minors. Conjectures 7.2, 4.2 and 4.4 of the same paper are proved in the preorder-polytope reciprocity report.

82. Largest Simplex Faces under Ordinal Sums

enumerative-combinatorics/

simplex-faces-of-order-and-chain-polytopes/Largest_Simplex_Faces.pdf

from Series_Parallel_Simplex_Faces_Research.zip

Answers the concluding open question of Freij-Hollanti and Lundström, *Simplex inequalities of order and chain polytopes of recursively defined posets* (Order 43 (2026), article 31). For $s_O(P)$ and $s_C(P)$, the largest dimensions of simplex *faces* of the order and chain polytopes: both are computable in $O(n)$ on any series-parallel decomposition, the order case carrying eight normalized endpoint states. For an ordinal sum of antichains, $s_O = r + \nu$ with ν a maximum b -matching of the layer path under capacities $\min(a_i - 1, 2)$, and $s_C = r + \max(0, k - 1)$. The extremal gap is sharp: $0 \leq s_C - s_O \leq \lfloor (n - 2)/3 \rfloor$, attained for every $n \geq 2$.

83. Skew partitions, continued fractions and connected components

enumerative-combinatorics/skew-partition-continued-fractions/article.pdf

from A225114_proof_bundle.zip

Proves the repeated-power continued-fraction formula recorded as conjectural in OEIS A225114, which counts skew partition diagrams with no empty row or column. Unique concatenation gives $A(q) = 1/(1 - P(q))$ for the edge-connected diagrams, and a boundary-path encoding followed by a weighted two-coloured Motzkin-path decomposition gives $P(q) = q/(1 - 2q - q^3/(1 - 2q^2 - q^5/(1 - 2q^3 - \dots)))$, whence $A(q) = 1/(2 - C(q))$ with $C = 1 + P$. Adds a four-variable refinement, a formula by height, width and component count, a sharp truncation bound, exponential asymptotics and a central limit theorem for the number of components. **Explicitly not a novelty claim:** a prior-art check found Pasechnik’s MathOverflow sketch of ten September 2024, and the entry credits Kurkov; the report says so first and credits both.

84. From mesh avoidance to Catalan inflation

enumerative-combinatorics/mesh-avoidance-catalan-inflation/article.pdf

from A289587_proof.zip, OEIS_A289587_proof.zip

Proves Scheuerle’s generating-function conjecture in OEIS A289587 for permutations avoiding 321 and the mesh pattern (12, 174), and the equal count for (12, 234) by inversion: with $D = x^4 - 2x^3 - 5x^2 - 2x + 1$, $P = x^4 + 4x^3 + 11x^2 + 10x + 3$ and $Q = x^2 + 5x + 3$, the generating function is $(P - Q\sqrt{D})/(8x(1 + x)^2)$. The headline structural result is the exact *occurrence count* $\text{occ} = \binom{f}{2} + \sum_U \binom{|U|}{2}$, of which avoidance is the case $\text{occ} = 0$. Merged from two reports that reached the identical generating function by the identical run-contraction substitution $u \mapsto u/(1 + xu)$; the merge keeps *both* refinements, because they refine the same sequence by different statistics — excedances and records — and are genuinely different theorems, agreeing only in total. The two local predicates they are built on disagree on 31 of the 321-avoiders of length at most six, the smallest being 12, and the merged article proves where they coincide rather than assuming it.

85. Prescribed coefficient-lattice minima

enumerative-combinatorics/coefficient-lattice-minima/article.pdf

from coefficient_lattice_minima_research.zip

Settles the unnumbered realization conjecture after Lemma 1 of O’Bryant, *On Nathanson’s Triangular Number Phenomenon* (arXiv:2506.20836): every nondecreasing list of even integers is the list of successive L^1 -minima of the coefficient lattice of some finite integer set. The witness is the mixed-radix set $A = \{0, 1, h_1, h_1 h_2, \dots\}$, whose minima are exactly $(2h_1, \dots, 2h_r)$ with $r + 2$ distinct elements, and the radices may be given in any order.

86. Coefficientwise magic positivity through dimension ten

enumerative-combinatorics/magic-positivity-parking-polytopes/article.pdf

from magic_positivity_research.zip

A polytope is *magic positive* when its Ehrhart polynomial has nonnegative coefficients in the basis $t^j(t + 1)^{n-j}$. Proves, in exact arithmetic, that the generalized parking-function polytope $X_n(b)$ is magic positive for every dimension $3 \leq n \leq 10$ and *every* positive integer parameter vector, uniformly in b rather than on a sampled box — a partial solution of Conjecture 8.1 of Hill, Luo,

Trinh and Vindas-Meléndez (arXiv:2607.15503v1). Dimension three is proved by printed algebraic formulas; the first-coefficient harmonic-number formula and the endpoint identities hold in every dimension. The conjecture in unrestricted dimension remains open.

87. Monotone blocks and reverse replies

[enumerative-combinatorics/reverse-reply-avoidance-game/article.pdf](#)
[from reverse_reply_solution.zip](#)

Proves Conjecture 2.9 of Ulfarsson (arXiv:2603.16004): for every $k \geq 3$ and every $n \geq R(k) = k^4 - 2k^3 - 3k^2 + 5k + 5$, the second player wins the normal-play permutation-avoidance game on S_n with pattern length k by always replying with the reversal, so the Sprague–Grundy value is 0. The proof is elementary, with no computer search: a minimal-support lemma isolates one pattern inside a family built from singletons plus one inflatable monotone interval, and inflation restores the original length so the reply is legal. $R(k)$ is asserted sufficient, not minimal.

88. Nonintegral tropical degree from two horizontal segments

[enumerative-combinatorics/tropical-hilbert-nonintegral-degree/article.pdf](#)
[from tropical_hilbert_counterexample.zip](#)

Computes the strict-witness tropical Hilbert function of Grigoriev (arXiv:2404.06440v2) exactly for a two-segment set: $\text{TH}_{V_d}(k) = k + \min\{k, \lceil k/d \rceil\} + 1$. Taking $d = 3$ gives tropical degree $4/3$ — not an integer — and a quasipolynomial of least period 3, so eventual polynomiality fails. Every rational in $(1, 2]$ is realized, denominators are unbounded, and for $d = p/q > 1$ in lowest terms the least eventual period is exactly p . Both bounds are proved by hand.

89. Finite-prefix corrections for polyomino growth

[enumerative-combinatorics/polyomino-growth-finite-prefix-corrections/article.pdf](#)
[from ProveIt_Polyomino_Research.zip](#); batch 40: [ProveIt_Spectral_Saturation.zip](#)

Continues the formal project `Combinatorics/Polyominoes/KlarnerConstant`, whose Lean-verified bound is $\lambda \leq 9047/2000 = 4.5235$ for Klarner’s constant. Proves a finite-prefix correction theorem for proper positive polynomial recurrences and applies it to Bui’s seventeen-neighbourhood system with an exact marked-occurrence table through size 18, giving the computer-assisted, unformalized bound $\lambda \leq 2249/500 = 4.498$ and $A_n \leq (927884613/10^9)(2249/500)^n$. A second theorem gives balanced integer-monomial divergence certificates, complete for productive, strongly connected, nonlinear systems, and yields $4.52349 \leq \mu_{\text{original}} \leq 4.5235$ for the original recurrence — a statement about the recurrence, not a lower bound on λ . At intake, a fresh build of the delivered enumerator (the same program, not an independent one) regenerated the size-18 table and matched it exactly. Part II, added in batch 40, shows where the method stops: exact spectral saturation in the irreducible critical case, a square-root law, and an all-prefix barrier $\mu_N \geq 4.3149$ for Bui’s map — a limit of this method, not a lower bound on λ and not an improvement on 4.498.

12 Generating functions and asymptotics

12.1 Coefficient counts of binary products

Three reports on counting coefficients of $\prod_i (1 + x^{G_i})$. The first two answer, independently, the universal-rationality part of Richard Stanley’s MathOverflow question 431075 (“A conjectured rational generating function”, 2022), which had no posted answer when checked on 19 September 2026. A third answer used the same mixed-base construction and the same enumerative argument as the first, and has been merged into it.

90. A Mixed-Base Counterexample to Stanley’s Rationality Question

[generating-functions-and-asymptotics/binary-product-coefficient-counts/mixed-base-counterexample/article.pdf](#)
[from stanley_mixed_base_counterexample.zip](#) + [Stanley_Rationality_Counterexample.zip](#)

With $G_{2j+1} = 2^j$ and $G_{2j+2} = 3^j$ the exponent generating function is rational, while $\nu(2n) = 2^{n - \lfloor (n+1)\alpha \rfloor}$ for $\alpha = \log_3 2$ counts the coefficients equal to one in $\prod_{i \leq N} (1 + x^{G_i})$ and its generating

function is not rational, not D-finite, not P-recursive, and has a natural boundary. Classifies rationality in the family interleaving a^j and q^j with digit polynomial $1 + y + \dots + y^{a-1}$ for all $2 \leq a < q$ — Stanley’s general $r = a - 1$ — as holding exactly when $q^u = a^v$. Adds a finite-quotient rigidity theorem, an abstract natural-boundary theorem for $\sum q^{\lfloor n\theta \rfloor} t^n$ at every irrational θ , the exact limit set of a_n/λ^n , and a streaming recurrence making the point that non-P-recursive does not mean hard to compute. *Merged; the enumerative half was the same argument in both originals, but the two routes to non-rationality differ and both are kept — an elementary finite-state reduction mod p , and the rigidity route that reaches the stronger transcendence conclusions.*

91. Rational exponents, nonrational coefficient counts

[generating-functions-and-asymptotics/binary-product-coefficient-counts/recursive-exponent-construction/research.pdf](#)
 from stanley_counterexample.zip

A different negative answer, from the recursion $a_0 = 2$, $a_1 = 1$, $a_n = a_{n-1} - 2a_{n-2}$, with $b_n = T_{n-1} + 2^n - 1 + a_n$, $T_n = T_{n-1} + b_n$, $G_{2n-1} = 2^{n-1}$ and $G_{2n} = b_n$: the counting series has a natural boundary on the unit circle, hence is neither rational, algebraic nor D-finite.

92. Coefficient residues in generalized Fibonacci products

[generating-functions-and-asymptotics/binary-product-coefficient-counts/kbonacci-coefficient-residues/article.pdf](#)
 from stanley_kbonacci_research.zip + quasifibonacci_parity_research.zip

Settles Conjectures 6.2 and 6.3 of Stanley, *Theorems and Conjectures on Some Rational Generating Functions* (arXiv:2101.02131v3, pp. 23–24). For k -bonacci seeds each exceeding the sum of their predecessors, the number of odd coefficients of $\prod_{i \leq n} (1 + x^{w_i})$ has generating function $(1 + 2z^k)/(1 - 2z + 2z^k - 2z^{k+1})$, independently of the seeds; the proof builds periodic signs of block length $k + 1$ and block product -1 and shows every signed finite product is flat. *Merged; the Conjecture 6.3 halves proved the same statement on the same domain, and the surviving report adds Conjecture 6.2 and the asymptotic analysis.*

12.2 Single-sequence OEIS studies

Nine reports on the growth or the closed form of a single OEIS sequence. Three of them count the terms in the derivatives of a power tower — of x^{x^x} , x^{x^2} and x^x — and reach three different answers by three different recurrences; two more share a Bell-scale saddle-point method on unrelated sequences. Shared technique is not shared theorem, and none of these nine duplicates another.

93. Bell-scale growth of OEIS A088714

[generating-functions-and-asymptotics/oeis-sequence-asymptotics/a088714-bell-scale-growth/article.pdf](#)
 from A088714_asymptotic_growth.zip

For the unique formal series with $A(x) = 1 + xA(x)^2A(xA(x))$, proves $\log a_n = n \log n - n \log \log n - n + O(n \log \log n / \log n)$, so $(e \log n / n) a_n^{1/n} \rightarrow 1$ and the ordinary generating function has radius of convergence zero while the exponential one is entire. The lower bound comes from a single explicit path through an exact triangular recurrence, the upper from a Stirling-positivity comparison with derivatives of Touchard polynomials. The multiplicative equivalent is *not* claimed, and the finer Bell-number form is labelled a conjecture with a sufficient ratio condition isolated.

94. The exponential growth constant of OEIS A158415

[generating-functions-and-asymptotics/oeis-sequence-asymptotics/a158415-growth-constant/article.pdf](#)
 from A158415_asymptotic_growth.zip

For the number of distinct values expressible with n symbols from $E ::= 1 \mid \sqrt{E} \mid (E + E)$, proves a computable $\gamma > 1$ exists with $a(n)^{1/n} \rightarrow \gamma$ and $1.86204 < \gamma < 1.9208$, gives an effective characterisation as a supremum of algebraic finite-data bounds, and proves the boundary identity $\sum a(n)\gamma^{-n} = \gamma^2$, which rules out a pure $C\gamma^n$ prefactor. The rigorous lower bounds rest on a

uniform injectivity theorem: any finite alphabet of representable values can serve as nested-radical digits, via an odd valuation at a prime above 2. No closed form for γ is claimed, and terms 22–28 get certified intervals rather than exact values.

95. Dyadic scaling rigidity of 3AP-free permutations

[generating-functions-and-asymptotics/oeis-sequence-asymptotics/a003407-dyadic-scaling-rigidity/article.pdf](#)

from ProveIt_Dyadic_Scaling_Rigidity.zip (batch 40); batch 43: ProveIt_3AP_Nonholonomicity.zip

For $\theta(n)$, the number of permutations of $[n]$ with no three-term arithmetic progression (OEIS A003407), studies the dyadic scaling of $\theta(n)$: a continuous log-periodic growth profile with its exact fundamental period, rigidity of the profile under real rescaling, explicit losses and gains of the 2:1 gluing, and the empirical and sampling laws; the adjacent ratios $\theta(n+1)/\theta(n)$ have no limit. The parity recurrences and Sharma’s Theorem 2.8, which the manuscript imports, are already proved in Lean in `Combinatorics/Ramsey` (the table $\theta(1..20)$ there rests on a `native_decide` certificate); the report’s own theorems are not formalized. Part II, added in batch 43, proves θ is not P-recursive and its generating functions are not D-finite (the ordinary one is transcendental), using the published counts $\theta(64)$, $\theta(75)$, which nothing in ProveIt certifies; whether the generating function has a natural boundary stays open.

96. An exact coefficient formula for Ramanujan’s theta function

[generating-functions-and-asymptotics/oeis-sequence-asymptotics/a260306-formula-proof/a260306_proof.pdf](#)

from A260306_formula_proof.zip

Proves both finite formulas recorded in OEIS A260306, for every coefficient index, for the expansion $\theta(N) \sim \sum c_n N^{-n}$ defined by $\sum_{k < N} N^k/k! + \theta(N)N^N/N! = e^N/2$. The route is an exact Laplace-integral difference, an analytic change of variable with $F(\psi(w)) = w^2/2$, and Lagrange inversion reducing each c_n to one coefficient extraction, then a finite change of basis into the entry’s double and triple sums. Includes the analytic error estimate the identification needs, rather than only checking the two sums agree, and identifies A065973 as the reduced denominators.

97. The maximal positive coefficient of a Hermite polynomial

[generating-functions-and-asymptotics/oeis-sequence-asymptotics/a277280-hermite-maximum/article.pdf](#)

from A277280_asymptotics.zip

An exact closed form for OEIS A277280, the largest coefficient of the physicists’ Hermite polynomial H_n with signs retained: with $k_n = \lfloor (2n+6 - \lfloor \sqrt{8n+21} \rfloor)/8 \rfloor$ and $d_n = n - 4k_n$, the exponent d_n is the unique maximizer and $a(n) = n! 2^{d_n} / ((2k_n)! d_n!)$. The distinction from the largest coefficient in absolute value is kept explicit throughout.

98. Asymptotics of OEIS A277364

[generating-functions-and-asymptotics/oeis-sequence-asymptotics/a277364-bell-asymptotics/A277364_asymptotics.pdf](#)

from A277364_asymptotics.zip

For $a(n) = \sum_{k \leq n/2} S(n, k)$, partitions of an n -set into at most $\lfloor n/2 \rfloor$ blocks, proves $a(n) \sim B_n$ with an explicit first correction: writing $r = W_0(n)$, $a(n) = \exp\{n(r-1+1/r)-1\}(1+r)^{-1/2}(1-P(r)/n+O(r^2/n^2))$ for an explicit rational P . Also gives a parity-split equivalent for the omitted tail $B_n - a(n)$.

99. Term counts in the derivatives of power towers

[generating-functions-and-asymptotics/oeis-sequence-asymptotics/power-tower-derivative-term-counts/article.pdf](#)

from A293239_research_report.zip, A290268_partial_results.zip, A281434_exact_algorithm_and_cubic_growth.zip; batch 42: ProveIt_A290268_Finite_Certificates.zip

One article on three OEIS sequences that ask the same question of three functions: how many distinct monomials are in the fully collected n th derivative of x^x (A293239), x^{x^2} (A290268) and x^{x^x} (A281434)? Part I proves once the apparatus all three had established separately —

the canonical differential-polynomial form $f^{(n)} = fP_n$ or $fx^{-n}P_n$ with its first-order integer recurrence, the independence of logarithmic monomials that makes the count an invariant of f rather than of a differentiation algorithm, the identity $a(n) = |\text{supp } P_n|$, and the envelope bound. Two normalizations are needed, because neither clause covers all three in the frames the parts use. **None of the three headline questions is fully settled, and the article says so in its abstract:** for x^x the count reduces exactly to zeros of the first-kind Lehmer–Comtet triangle A008296 and $a(n) = \Theta(n^2)$, but the closed formula needs an isolated nonvanishing step — and separately the recurrence printed in the entry for $n > 6$ is *disproved* at $n = 8$; for x^{x^2} the conjectured expression is proved to be an upper bound and $a(n) = \Theta(n^2)$, but not the constant $\frac{1}{2}$; for x^{x^x} explicit bounds give $a(n) = \Theta(n^3)$ but not $a(n) \sim \frac{2}{3}n^3$, and there the obstruction is not a nonvanishing statement but a density one, since further zeros provably exist. A closing part sets the three side by side and records what they share beyond the framework: every cancellation any of them can prove is a parity zero of the same falling factorial. This is a thematic consolidation, not a deduplication: the three were checked and are not duplicates, and all fifty numbered results of the three originals are preserved unchanged. Batch 42 added to the $x^{(x^2)}$ part depth certificates that close the nonvanishing problem at logarithmic deficits 3 and 4 for every n and improve its lower bound on the count; the open region is now deficit at least 5. Where that manuscript re-derived results already in the report, it is printed as a second route.

100. Iterated sums and products from one

[generating-functions-and-asymptotics/oeis-sequence-asymptotics/a352969-correctness-and-growth/article.pdf](#)
from A352969_correctness_and_growth.zip

For $S_0 = \{1\}$, $S_{n+1} = (S_n + S_n) \cup (S_n \cdot S_n)$ and $a_n = |S_n|$ (OEIS A352969), proves correctness of the listed terms and the doubly exponential law $2^{2^{n-rn-4}} \leq a_n \leq 2^{2^{n-1}}$, hence $\log \log a_n = n + O(\sqrt{\log n})$. The step is simultaneous, not repeated closure — a distinction the report is careful about.

101. Distinct third derivatives of a power tower

[generating-functions-and-asymptotics/oeis-sequence-asymptotics/a199085-exponentiation-derivative-values/article.pdf](#)
from A199085_generating_function.zip

For OEIS A199085, the number of distinct values taken by the third derivative at $x = 1$ over all fully parenthesized exponentiations of n copies of x , proves $a(1) = a(2) = 1$ and $a(n) = \lfloor (n^2 - 2)/3 \rfloor$ for $n \geq 3$, with the rational generating function $z/(1-z) + z^3/(1-z)^3 - z^7/((1-z)^2(1-z^3))$. The proof is for all n , not an extrapolation: every attainable pair of second and third derivatives is classified and a witness constructed for each admissible pair. Despite the shared vocabulary this is a different question from *power-tower-derivative-term-counts*, which fixes the function and varies the derivative order; here the order is fixed at three and the parenthesization varies.

12.3 Asymptotics, records and discrepancy

102. The Exponential Scale of Gregory Sign Thresholds

[generating-functions-and-asymptotics/gregory-coefficient-sign-thresholds/gregory_sign_threshold.pdf](#)
from gregory_sign_threshold_research.zip

For $(x/\log(1+x))^m = \sum_n G_n^{(m)} x^n$ and $\text{Gr}(m)$ the least N beyond which $(-1)^{n-1} G_n^{(m)} > 0$, proves $\log \text{Gr}(m) = m + 1 - \gamma - (\zeta(2) + \zeta(3))/m + C_2/m^2 + O(m^{-3})$ with C_2 explicit, plus a finite algorithm for every coefficient of the complete expansion; in particular $\text{Gr}(m)/e^m \rightarrow e^{1-\gamma}$ and $\text{Gr}(m)/3^m \rightarrow 0$. This replaces Conjecture 6.1 of Xu and Zhao (arXiv:2609.11072v1), whose printed lower bound already conflicts with its own reported $\text{Gr}(5) = 113$; that value is independently certified here.

103. Exact Records in the Square-Root-of-Three Walk

[generating-functions-and-asymptotics/sqrt-three-walk-records/article.pdf](#)
from [sqrt3_record_recurrence.zip](#)

Proves the four-phase record recurrence for $S(N) = \sum_{n \leq N} (-1)^{\lfloor n\sqrt{3} \rfloor}$ in the corrected form on page 7 of Bruin and Fokkink, *On the records and zeros of a deterministic random walk* (arXiv:2503.11734v2), where it is experimental and attributed to Arias de Reyna and van de Lune. Includes minimality of every claimed first passage, a rational generating function, closed forms, sharp logarithmic extrema, and an extension to every even integer trace $D \geq 4$.

104. Sharp discrepancy for a family of binary substitutions

[generating-functions-and-asymptotics/binary-substitution-discrepancy/article.pdf](#)
from [oeis_discrepancy_counterexamples.zip](#)

For the fixed word of $0 \mapsto 1, 1 \mapsto 101010$, the position sequences A284365 and A284366 both carry a conjectured upper position error below 2, and both conjectures are false: the first counterexamples are at ranks 2,977,771 and 8,933,313, with common error $(2977771\sqrt{21} - 13645857)/2 \approx 2.00487$ and sharp common upper bound $(6 + \sqrt{21})/5 \approx 2.11652$. Proves complete discrepancy limit intervals for every substitution $0 \mapsto 1, 1 \mapsto (10)^m$, with infinite counterexample families and exact recurrences for their indices.

105. Exact eventual maximizers of Rudin–Shapiro autocorrelations

[generating-functions-and-asymptotics/rudin-shapiro-autocorrelation-maximizers/article.pdf](#)
from [rudin_shapiro_maximizers.zip](#)

For the aperiodic autocorrelation $C_m(k)$ of the level- m Rudin–Shapiro sequence, classifies the maximizing shift exactly for all large m : $k_m^* = (2^{m+1} + (-1)^m)/3$, with $A_m = b_{m-2}$, settling the discrete uniqueness-and-location conjecture of Tarnu (J. Approx. Theory 287 (2023) 105866), repeated as Conjecture 1.5 of Choi–Tarnu (2025), including the limit $k_m^*/2^m \rightarrow 2/3$. The certificate is a rational polytope argument with exact arithmetic.

106. Apéry arrays and conjectural zeta accelerations

[generating-functions-and-asymptotics/apery-array-zeta-accelerations/article.pdf](#)
from [Apery_OEIS_Conjectures_Proofs.zip](#), [apery_oeis_conjectures.zip](#)

Proves the entire shifted-diagonal $\zeta(3)$ family recorded as conjectural in OEIS A143007, and both the super- and subdiagonal $\zeta(2)$ families of A108625, at every nonnegative offset: for all $k \geq 0$, $\zeta(3) = H_k^{(3)} + \sum_{n \geq 1} (2n+k)(3n^2+3nk+k^2)/(n^3(n+k)^3 T_{n-1,n+k-1} T_{n,n+k})$. The main diagonals are the classical Apéry sequences A005259 and A005258. Proofs are finite telescoping identities with rational lattice potentials and explicit limit estimates; no computer test or unproved conjecture enters. Adds rational error bounds, numerator and denominator recurrences, a three-term recurrence along every shifted diagonal, and sharp fixed-offset asymptotic constants. Merged from two reports whose boxed identities were character-identical up to macro names.

13 The Jacobian conjecture and Keller maps

Three reports continuing the formal project [Algebra/JacobianConjecture](#), which verifies in Lean and Rocq Alpöge’s three-variable counterexample and a degree-six stable representative in five variables (its stabilization to all higher dimensions in Lean only); its further degree reductions are SymPy certificates. None of the three reports’ own theorems is formalized; each README names the project declarations it builds on. Two study the project’s three-variable map; the third studies a different map, Gao’s five-dimensional F_6 .

107. The fiber geometry of Gao’s Keller map F_6

[jacobian-conjecture/gao-f6-fiber-geometry/article.pdf](#)
from [ProveIt_Six_Sheets_Research.zip](#), [ProveIt_Keller_Fibers_Research.zip](#)

For Gao’s five-dimensional six-sheeted Keller map F_6 (arXiv:2608.00222v1, §4.5.1), classifies every geometric fiber: the cardinalities are exactly 0, 1, 2, 3, 4, 6, all seven root-multiplicity patterns

occur, and off the hyperplane $c = 0$ the inverse points are the simple roots of an explicit sextic. The omitted set is a smooth closed surface isomorphic to $\mathbb{G}_m \times \mathbb{A}^1$, of codimension three, with the single effective omitted parameter $(21/32, 5/32, 25/128)$. Gives an explicit finite normalization of rank six, the two-component nonproperness locus, the discriminant's normalization and local analytic types, global monodromy S_6 with two arithmetic certificates, no proper intermediate field, real fiber cardinalities $\{0, 1, 2, 3, 4\}$, and an obstruction to equivalence with a stabilization of a map in at most three variables. The map is Gao's; the fiber theory is the reports'. Merged from two manuscripts proving the same core in identical normalizations; the second one's different proofs are kept as marked second routes. Not to be confused with the project's formally verified dimension-five stable map.

108. Arithmetic local-global dichotomies for the three-variable Keller map

[jacobian-conjecture/arithmetic-local-global-fibers/article.pdf](#)

from ProveIt_Arithmetic_Local_Global.zip

For the project's three-variable Keller map, gives an explicit infinite family of integral targets with a preimage in every $\mathbb{Z}_p^3$ but none in $\mathbb{Z}^3$ (each fiber is three rational points, so these are not rational Hasse failures); classifies the split integer-root plane $C = 2$ by a gcd condition, with locally soluble parameter density $27/(4\pi^2)$ surviving the removal of S -integral fibers; proves the exact integral 2-adic fiber law $(21, 7, 3, 1)/32$ for fiber sizes 0, 1, 2, 3, determined modulo 8; and contrasts it with a rational Hasse principle over every number field. The complex stratification is credited to Gao and the finite-field histograms to an external note. An intake remark records an integral triple collision over the project's own collision value. Not a new disproof of the Jacobian conjecture.

109. All-degree rigidity of a weighted Keller class

[jacobian-conjecture/weighted-keller-rigidity/article.pdf](#)

from ProveIt_Weighted_Keller_Rigidity.zip; batch 44:

ProveIt_Affine_Weighted_Keller_Classification.zip

For polynomial maps of weights $(-1, 1, 2)$ written through $t = xy$ and $v = x^2z$, classifies the Keller maps in every degree over any field of characteristic zero: those with p, q affine in v and r affine in (t, v) are an explicit tame family and a two-parameter diagonal orbit of the known map, so every noninvertible member has coordinate degrees $(7, 6, 4)$ and sixteen monomials, with no input degree bound. Extends to $r = R(t) + \gamma v$, where the possible maximum degrees of the noninvertible members are exactly 7 and every even integer at least 8, and shows that with $r = 1 + t + v$ the fixed degree-seven coefficient scheme is the double point $\mathbb{Q}[\varepsilon]/(\varepsilon^2)$, with an explicitly obstructed tangent (a torus times it when the two coefficients of r are freed). The classification does not cover unrestricted weighted maps. Sharpens to all degrees the project's degree-seven sparsity search; it does not show degree seven minimal in dimension three. Part II, added in batch 44, removes the restriction on the coefficient of v in r : whenever p, q, r are affine in v that coefficient is constant, so the classification covers the whole affine-in- z class; it adds dual-number slopes of every degree and a universal second-order obstruction.

14 Galois theory and radicals

Two reports continuing the formal project `Algebra/PolynomialFormulas` (Lean and Rocq root formulas through degree four, Abel–Ruffini, and Lazard's quintic solver); none of its theorems is formalized.

110. Specialization-safe radical solvers

[galois-theory-and-radicals/specialization-safe-radical-solvers/article.pdf](#)

from ProveIt_Radical_Solvers_Research.zip; batch 52: ProveIt_Optimal_Kummer_Atlases.zip

Gives an exact cyclotomic-resultant criterion for the characteristics in which equal-cardinality subset sums of the roots of an irreducible prime-degree polynomial can collide, and evaluates it: the bad characteristics are $\{5\}$ for quintic pairs, $\{2, 7\}$ for septic pairs and $\{2, 7, 29\}$ for septic triples. Classifies the collisions completely in characteristic 29 (exactly the cyclic polynomials

$(X - \eta)^7 - b)$ and in characteristic 2 (translated $Z^7 + aZ^3 + bZ + d$, realizing the nonsolvable $\mathrm{GL}_3(\mathbb{F}_2)$). Constructs separating subset invariants in every characteristic, reconstructs all roots from one coherent p -th radical on each Fourier chart once the intermediate-field invariant data are given (generalizing Lazard’s pivot-one scheme for quintics; a coefficient-only solver is left as further work), realizes every Fourier support by rational polynomials with full affine Galois group, and shows the Picard group of the cyclic configuration quotient is $\mathbb{Z}/p$, so no single global Kummer coordinate exists. The Picard result has a written proof only. Part II, added in batch 52, answers Part I’s question on the least number of charts: $\mathrm{CH}^*(\mathrm{Conf}_p(\mathbb{A}^d)/C_p) = \mathbb{Z}[t]/(pt, t^{d(p-1)})$, so every cover by regular Kummer charts, even with nonlinear character sections and different characters on different charts, needs exactly $d(p-1)$ charts ($p-1$ for polynomials: four for quintics, six for septics), and the Fourier charts attain it. This too is a written proof, by integral intersection theory; its checker verifies only finite identities.

111. Finite separating ranges for sextic block resolvents

[galois-theory-and-radicals/sextic-block-resolvent-separators/article.pdf](#)

from `ProveIt_Finite_Sextic_Separators.zip` (batch 41)

The project finds parameters separating the sextic pair and triple block-resolvent descriptors by an unbounded `Nat.find` search. This report proves in writing that on the curve $(t, 2t^2)$ one of $t = 1, \dots, 196$ always separates both families, sharply for arbitrary complex test sets, by orbit counting. The bound is not formalized; formalizing it would make that search bounded, and two further unbounded searches of the project remain.